

Language as Visible Signal: Toward an Image-Native Language Model in Continuous State and Pixel Space

Imagized Language Model Project

August 2026

Abstract

Most language models treat language as a sequence from a finite token and Unicode-accessible inventory, even when an image encoder or image generator is attached. That interface is incomplete for historical, regional, handwritten, damaged, undeciphered, and newly formed writing, and it discards visual form that can itself carry linguistic and historical evidence. OCR can transcribe the encodable subset but does not make writing the model’s language substrate. We define a stricter target: an independent Imagized Language Model (ILM) whose inputs, learned states, predictive random variables, and primary outputs are continuous images of writing. Transcription is one supported task, alongside continuation, translation, explanation, comparison, and generation; it is not the architecture.

We contribute a staged, falsifiable implementation. Retinal Flow Language Models read ordered 32×32 fixations, integrate visual history, generate the next ink field, and reread their own pixels, but expose the failure of a monolithic language-and-typography writer. Predictive Visual Fields (PVF) factorize the problem: a convolutional retina maps writing to continuous state, a causal field integrates history, a deterministic proposal predicts the next state, an intrinsic hyperspherical flow models alternatives, and a separate motor planner renders pixels. The learned path has no tokenizer, strings after rasterization, Unicode or character IDs, OCR transcript, output vocabulary, discrete visual codebook, glyph lookup, candidate classifier, or external language model.

On a frozen 512-form, four-view Chinese benchmark, the 16.47M-parameter V16 proposal reaches 6.26% top-1, above last-image-only (3.97%), unigram (1.73%), and random dynamics (0.22%); full history adds +0.0773 normalized target log-probability. Its stochastic field reaches 3.69% versus 3.24% last-only. V16 trains on one RTX 4090 at 1.479 GiB peak allocated CUDA memory, but remains below a symbolic bigram (13.14%). A separate 5.73M-parameter V17 actuator proves causal state control on an untouched evaluation—58.59% visual identity versus 0.98% after shuffling intended state—but fails readability. A topology-first 2.36M-parameter V18 planner reaches 73.63% identity top-1 versus 0.98% shuffled and pixel F1 0.658 versus 0.313 on a fresh development audit at 0.778 GiB peak memory. Simple and medium forms become recognizable; dense forms remain malformed, and V18’s frozen bank remains sealed. V19 then tests a zero-initialized spatial-retinal residual under fixed complexity strata and five causal interventions. Its fresh 512-candidate development audit rejects the mechanism: dense F1 is 0.728, but correct fields improve it by only 0.009 over shuffled fields and 0.005 over zero fields, against fixed margins of 0.12 and 0.03. V20 then constrains the global route to coarse 4×4 block logits and reserves within-block detail for a continuous local retinal field, with an exactly parameter-matched global-repeat control. The field becomes causally necessary: dense F1 is 0.701, versus 0.580 shuffled and 0.340 zeroed, and quadrant-occlusion locality is 1.0. The writer remains rejected because overall F1 is 0.636, target cosine is 0.815, the exact-decomposition tolerance fails, and final dense gain over the control is only 0.018 against a fixed

0.03 margin. Neither arm selects; no human or frozen query is authorized. V21 removes the remaining global spatial route: each local field cell emits coarse occupancy and 63 coefficients in an exact zero-DC patch basis, while global state supplies only spatially uniform modulation. At its best diagnostic snapshot, dense F1 is 0.705 versus 0.531 shuffled-field and 0.359 zero-field; identity top-1 is 79.10%, target cosine is 0.833, and locality and every algebraic invariant pass. The exact-parameter tiled-global control collapses to repeated patches. V21 still rejects the writer because overall, simple, and medium F1 (0.604, 0.565, and 0.594) miss fixed gates. No candidate selects, so the paired, human, and frozen stages remain forbidden. V22 then implements a six-frame visual prompt and one-frame image answer with two exactly matched 3.410M-parameter arms. A query-aware candidate sees the final visual query; a query-blind control receives a continuous null state. The candidate reaches only 0.78% counterfactual switch accuracy versus 0% control, 15.92% identity top-1 versus 15.33%, and pixel F1 0.511 versus 0.513. A development-only audit finds that its selector argmax is the operation frame in all 1,024 original and counterfactual prompts, while mean query attention is 1.37×10^{-13} . V22 therefore rejects single-frame selection as a relational binding mechanism. No candidate selects and frozen images remain uninstantiated. V23 replaces single-frame selection with relation before generation. A separately selected 1.122M-parameter image canonicalizer is frozen; a 25,602-parameter relation circuit compares query and label images, reads the operation image, routes one of two visible source glyphs, and emits a canonical answer image. On a fresh 1,024-episode development audit, the candidate reaches 99.80% query and operation switch accuracy and 99.51% identity top-1. Exact query-blind and operation-blind controls have zero switch and zero pixel change for their hidden factor. An opaque agent visual review scores 48/48 overall and 12/12 held-out. The single authorized frozen evaluation over 98 unseen identities reaches 99.83% binary choice, 99.61% query switch, 99.71% operation switch, 99.46% identity top-1, and 0.785 pixel F1. Every fixed gate passes without frozen model selection or threshold changes. This accepts a fixed two-pair same/other visual grammar, not free-form language. V24 removes absolute frame roles and fixed input length while closing a two-frame autoregressive visual loop. Visible header images locate roles in 15–24-frame packet streams; frame 1 renders the selected unseen glyph, and frame 2 is produced after rereading the actual generated frame-1 pixels. Only 1,347 parameters are trained. Fresh paired query-, operation-, history-, and header-blind controls localize every required cause. An opaque agent review scores 47/48 for frame 1 and 48/48 for frame 2. The single authorized frozen run over 107 unseen identities reaches 98.67% frame-1 identity, 99.73% frame-2 label accuracy, 97.66% query switch, 97.27% operation switch, and 99.61% history switch. Every fixed gate passes. This accepts a variable-input but fixed-grammar two-frame image stream, not arbitrary rendered language. V25 removes that grammar and trains a 25.55M-parameter causal field on ordinary Chinese book streams represented as 64 visible 32×32 cells. On 2,048 fixed development windows, full-history top-1 is 1.123%, versus 0.146% last-only, 0.342% after shuffling prior history, 1.611% image unigram, and 12.158% symbolic bigram. The ordered-history effect is measurable but six fixed semantic and causal gates fail. Peak allocated CUDA memory is 0.598 GiB, the student boundary passes, and the frozen partition remains sealed. An explicitly exploratory writer trained after rejection preserves ink density but obtains zero generated-identity top-1, reread cosine 0.080, and pixel F1 0.322. V25 therefore rejects this natural-language objective and writer rather than treating low memory or glyph-like marks as language efficiency. V26 tests the proposed repair with a 19.14M-parameter factorized model: the last glyph supplies appearance, the preceding 63 glyph images supply a separate history residual, and a continuous head predicts eight visual particles at future offsets 1, 2, 4, and 8. After 8,000 fixed BF16 updates, full-history top-1 is 0.0488%, below 1.4160% image unigram and 13.5254% symbolic bigram. For 512 cross-record pairs with pixel-identical four-glyph suffixes, history residuals differ by 4.628 on average, yet correct pair ranking and residual-swap target accuracy are both exactly 50%. A perfect retina-bank oracle rules out a blind evaluator. V26 therefore rejects the factorized mechanism: visual history changes state without binding that state to the correct next-glyph distribution. It uses 0.888 GiB peak allocated CUDA memory; no writer or frozen evaluation is opened. A post-hoc localization freezes every V26 parameter and trains a three-branch, 1.11M-parameter candidate-conditioned

visual probe for one pass over the 16,384 training pairs. On 512 disjoint-record development pairs, frozen history and fused state reach only 50.684% and 50.342% cross-font assignment accuracy, respectively, while raw-retina target identity reaches 99.951%. This diagnostic is not preregistered evidence, but it rules against preserving V26’s context encoder and replacing only its stochastic proposal. V27 then performs the preregistered joint test with an 18.60M-parameter V16-initialized retina, eight-layer causal context field, and deterministic arbitrary-candidate image compatibility. The fixed 8,000-update run reaches 1.611% natural top-1, below 2.051% image unigram and 12.598% symbolic bigram. On pixel-identical suffix pairs, full-context assignment is 50.708%, versus 50.562% after shuffling the earlier prefix. The untouched V16 retina still identifies the correct one of two cross-font candidates at 99.951% and candidate permutation error is zero. Separately, learned 1,024-bank identity is 94.873% and misses its 99% gate; these differently sized retrieval tests are not a before/after comparison. V27 therefore rejects the global context-query/candidate-key mechanism and preserves the frozen seal. V28 freezes the raw V16 retina and replaces one endpoint query with dense continuous visual-future energies at horizons 1, 2, and 4. Its 17.86M-parameter system completes 10,000 fixed BF16 updates and audit in 118.91 minutes with 1.144 GiB peak allocated CUDA memory. On the same-scope 1,024-way identity audit, an EMA semantic route improves from 92.041% raw retina identity to 96.436%. Full context also improves target log probability by 0.215 nat over a suffix-preserving prefix shuffle. These effects do not produce useful language choice: natural top-1 is 1.416%, below 1.855% image unigram and 13.135% symbolic bigram, while pixel-identical-suffix pair assignment is 49.561% versus 49.951% after prefix shuffling. V28 passes ten of fourteen mechanism gates and two of six language gates, so the mechanism is rejected without opening frozen images or training a writer. V29 executes the candidate-conditioned density-ratio test with a 20.08M- parameter image-only critic. An arbitrary candidate image cross-attends to all retained context states, while exact-suffix scoring defines a centered full-minus-suffix increment. Full target log probability gains 0.179 nat over suffix and 3.102 nat over shuffled-prefix context, but natural top-1 is only 2.344% versus 13.867% for a symbolic bigram. On 512 exact-suffix pairs, incremental assignment is 50.708% versus 49.561% shuffled and both-correct is 8.984%. The common suffix baseline cancels exactly in aggregate two-by-two assignment margin. V29 passes eight of fourteen mechanism gates and two of six language gates and is rejected without frozen evaluation or writer training.

The bounded product target is a Visual Word-Origin Book: rendered English or Chinese questions produce answer-page images containing readable explanation and provenance-linked historical or unencoded forms. The present experiments establish causal visual-language signal, compact continuous-image writing, a bounded image relation, and a bounded causal two-frame image stream, not autonomous question answering, broad readability, Qwen parity, or efficiency over text models. More generally, ILM treats writing as the first instance of language modeled as a continuous sensory field. A proposed Visual Language Stream adds sequence/time and optional geometric depth to the same image-native contract, so a page, book stream, 3D glyph string, or character movie differs by field shape rather than vocabulary. Only the 2D writing case is tested here; 3D, motion, speech, and vocal fields remain future hypotheses.

1 Introduction: Writing as Language

1.1 Pain point

Human readers encounter language as marks on pages, screens, stone, bone, metal, and moving hands. Those marks vary with writer, era, medium, damage, typography, and reading direction. A finite token or Unicode interface is therefore not a generally accessible inventory of written language: it is an extremely useful codec for a documented subset. Historical and regional forms may have no code point, two visually different forms may share one code point, and an OCR transcript may erase the form whose evolution is the question.

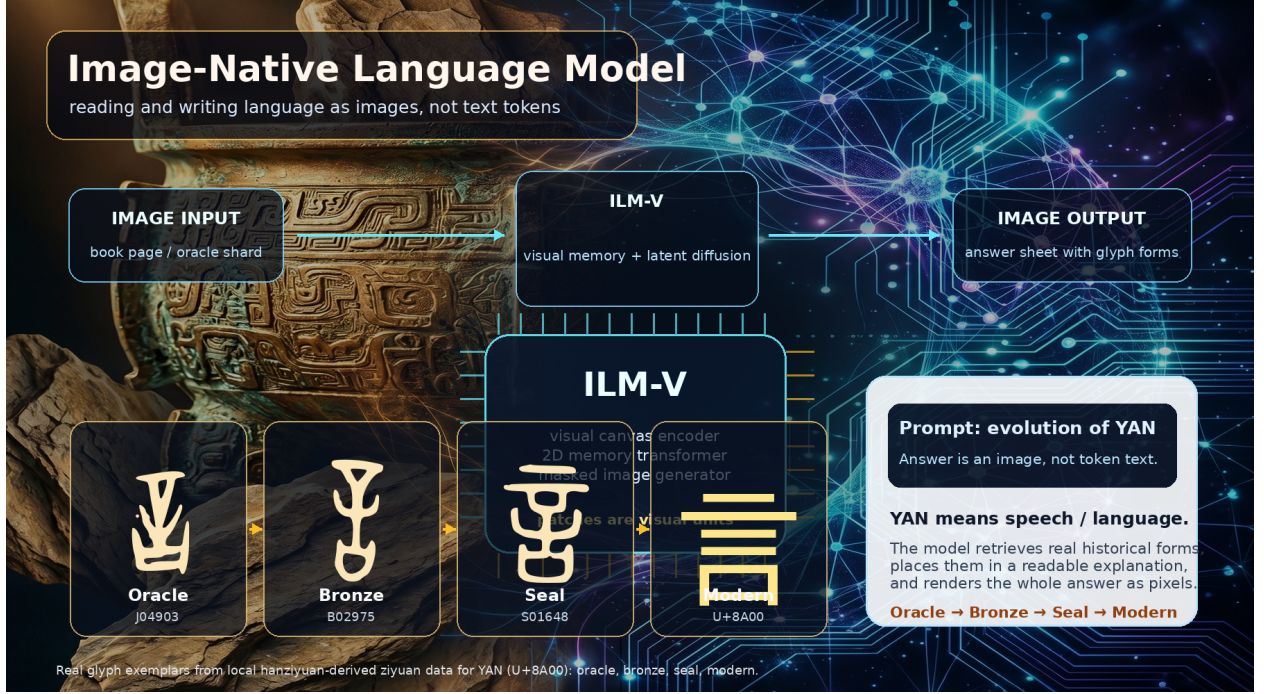


Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for YAN (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

Current language models are extremely capable, but their central language state is normally a sequence of symbolic tokens. Multimodal models can accept images, and modern APIs expose image generation [4, 5, 6]. These systems are important baselines, but attaching vision to a token-centered model does not make visible writing the predictive language substrate. An OCR system similarly solves a useful projection from image to available codes; it does not model all of the visual language that was projected away.

1.2 Goal and scope

The Imagized Language Model (ILM) instead treats written form as language:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output}.$$

The output should itself be a readable answer image. For a query such as “explain the evolution of character yan”, the system should produce a page containing modern explanation and attested oracle, bronze, seal, and modern forms, including forms unavailable in computer fonts. Typed input is rasterized before the core. A photographed page can be transcribed, but the same model should also continue, translate, explain, compare, and generate writing. Its answer is an image first; coded transcription is an optional downstream view.

The canonical input/output can be written as a Visual Language Stream,

$$X \in [0, 1]^{T \times D \times H \times W \times C}, \quad p_{\theta}(X_{t+1:t+k} \mid X_{\leq t}),$$

where T is reading sequence or visual time, D is optional geometric depth, H, W are spatial extent, and C is the sensor channel. A flat prompt has $T = 1, D = 1$; an ordered line or book has

$T > 1, D = 1$; a 3D glyph string has $D > 1$; and a 3D character movie uses depth and time. Collapsed axes have length one, so these are shapes of one continuous field rather than separate vocabularies. The present proof fixes $D = 1$ and short T ; depth and motion must earn separate causal and quality evidence after the 2D loop passes.

Prompt following partitions this stream into observed prompt frames and generated answer frames,

$$X_{\text{prompt}} \in [0, 1]^{T_p \times D \times H \times W \times C} \longrightarrow Y_{\text{answer}} \in [0, 1]^{T_a \times D \times H \times W \times C}.$$

A typed question is deterministically rendered into prompt frames; a scan or inscription enters natively. $T_a = 1$ is an answer page and $T_a > 1$ is an ordered text-image stream or movie. A language claim requires correct held-out changes in the generated answer when the prompt meaning changes. Input reconstruction, OCR accuracy, glyph classification, or attractive writing alone is insufficient.

The design is inspired by foveated visual reading, without claiming a biological simulation. Its broader hypothesis is sensory: the same perception–memory–prediction–motor–self-perception loop might later operate on speech waveforms or time-frequency fields. This paper tests visible writing only.

1.3 Contributions

The contribution is not the slogan “text as pixels.” It is a sequence of implemented controls that makes that slogan falsifiable:

1. A strict student boundary in which strings, token and Unicode IDs, OCR, finite visual codebooks, glyph lookup, and external language models are absent after rasterization and at deployed inference.
2. RFLM, a complete visual reading–prediction–writing–rereading loop that exposes autonomous generated pixels during evaluation and records its own negative result: stable ink is not readable language.
3. PVF, a mathematical factorization of continuous visual language state, causal memory, hyperspherical future-state uncertainty, and pixel actuation; V16 supplies controlled evidence that full visual history carries predictive language signal, while the symbolic bigram remains stronger.
4. Causal writer interventions: V17 proves that continuous image-derived intent controls generated identity, and V18 shows on development data that a 2.36M topology-first planner can render recognizable writing without token unembedding. Their failures delimit identity, topology, and language choice.
5. A preregistered V19 spatial-field intervention whose negative result separates feature availability from causal use, followed by an exactly capacity-matched V20 intervention that makes local topology necessary while rejecting its own writer-quality claim, and V21, which makes the field carry the complete spatial plan while identifying disjoint local raster patches as the next quality bottleneck.
6. V22, a strict six-frame image-prompt to image-answer experiment with a parameter-matched query-blind control. Its negative result localizes failure to relational binding: the selector collapses onto the operation image while writer quality and identity rise almost identically in both arms.

7. V23, a compositional visual relation circuit with an independently selected and frozen image canonicalizer, matched query-blind and operation-blind controls, held-out relation compositions, an opaque visual review, and one frozen-identity evaluation. It supplies the first accepted bounded image-prompt-to-image-answer result in this project.
8. V24, a variable 15–24-frame visual packet stream with learned header-image role localization and two autonomous output frames. Matched header-, query-, operation-, and generated-history-blind controls, an opaque visual audit, and one frozen run establish that frame 2 causally depends on rereading generated frame-1 pixels.
9. V25, a preregistered natural-Chinese visual-cell stream with fixed last-only, shuffled, blank, visual-frequency, text-bigram, and paired-history controls. It measures a weak ordered-history effect while rejecting the model under six fixed gates and preserving the frozen seal. A separately labeled writer diagnostic localizes stable ink without correct generated identity.
10. V26, a preregistered factorization of exact last-glyph appearance, earlier-history residual, and continuous future particles. Pixel-identical suffix pairs prove that history changes the residual state while exact-chance pair ranking proves that the state is not bound to the correct target. A perfect visual oracle separates this failure from retinal identity loss.
11. V27, a preregistered joint context/candidate image scorer with independently permuted candidate order, raw and learned cross-font identity controls, pixel-identical suffix pairs, and natural frequency baselines. It shows that preserving candidate visibility is insufficient: the learned global compatibility remains at chance and does not use prefix order.
12. V28, a preregistered dense multi-horizon visual-future energy model with a frozen raw retina, identity-initialized semantic residual, EMA target, and explicit suffix-preserving order losses. It improves same-scope visual identity and detects prefix information, while its negative rank and frequency-control results distinguish order sensitivity from correct conditional language binding.
13. V29, a preregistered candidate-conditioned visual density-ratio critic with exact shared-suffix scoring, full-minus-suffix contrast, independently shuffled prefixes, and candidate-permutation replay. It finds a large natural order effect but chance matched binding, and exposes the exact cancellation of a common suffix baseline in aggregate two-by-two assignment margin.
14. A rights-aware Visual Word-Origin Book benchmark and a continuous Visual Language Stream interface. They cover readable explanation, historical evidence, unencoded forms, optional depth and time, provenance, and one-GPU compute. The higher-dimensional interface is prospective.

2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. PIXEL reconstructs masked rendered-text patches and demonstrates cross-script visual transfer [10]; PIXAR is a genuinely generative pixel-space autoregressive language model and identifies noisy maximum-likelihood text images as a central difficulty [11]; PixelGPT studies autoregressive visual and textual pretraining [12]. These are the closest model-level precedents. Token-free language models such as CANINE, ByT5, and MEGABYTE remove subwords but retain symbolic bytes or characters. BLT and Scratchpad Patching allocate computation dynamically to high-entropy byte regions [16, 17];

ILM borrows the adaptive-compute question but not the byte substrate. Discrete diffusion language models show that generation need not be left-to-right, but usually retain a linguistic vocabulary.

Recent continuous approaches sharpen the distinction. Continuous visual autoregression can be trained with strictly proper energy scores without vector quantization [32]. VL-JEPA predicts continuous target-text embeddings but invokes a text decoder when output is required [33]. TextLDM reports that reconstruction-quality continuous latents are insufficient without alignment to a frozen pretrained language model [34]. These results motivate calibrated visual uncertainty and warn against equating reconstruction with semantics. Their byte, token, or text-decoder paths remain outside the independent ILM student boundary.

I-JEPA shows that predicting target image representations from visual context can learn semantic visual fields without reconstructing every pixel [28]. Flow matching trains continuous generative paths by vector-field regression and supports efficient numerical sampling [25]. Human reading is also sequential and foveated: Chinese readers make saccades and can use both character- and word-related targeting cues [38]. These observations motivate RFLM, but they do not imply biological equivalence.

The remaining gap is not simply “pixels instead of tokens.” It is a system whose predictive random variables are continuous writing images, whose memory is learned from ordered visual observations, whose writer produces new pixels, and whose own output distribution is present during evaluation. Historical evidence adds a separate constraint: synthesized marks must never be presented as attested source glyphs.

3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0, 1]^{H \times W}$, RFLM factors a visual sequence as

$$p_{\Theta}(x_{1:T}) = p(x_1) \prod_{t=1}^{T-1} p_{\Theta}(x_{t+1} \mid x_{\leq t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

Foveal retina. A convolutional reader maps each 32×32 image to a normalized 192-dimensional visual state:

$$z_t = R_{\theta}(x_t) / \|R_{\theta}(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\hat{\theta}}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

Recurrent visual field and energy. A three-layer GRU integrates the fixation history,

$$h_t = G_{\phi}(h_{t-1}, z_t), \quad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image c_j after rereading it:

$$s_{t,j} = E_{\psi}(h_t, z_t, R_{\hat{\theta}}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

Retinal Flow Language Model

Read visible fixations, predict a visual distribution, write ink, then read the model's own ink.

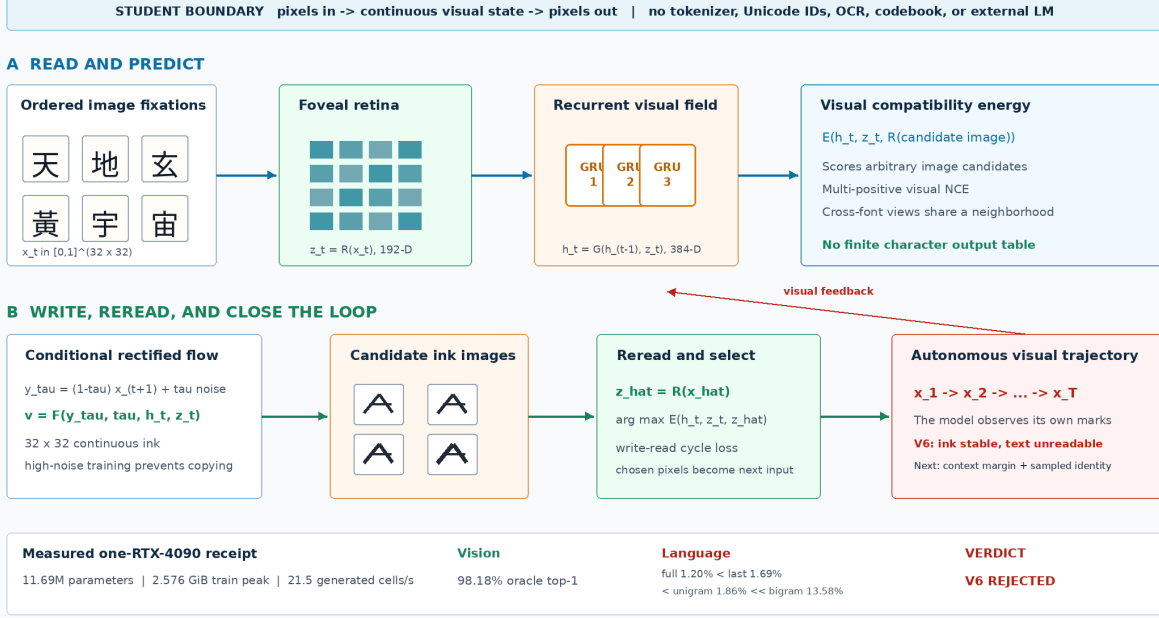


Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write–read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

Conditional ink flow. For clean target x_{t+1} , noise ϵ , and $\tau \in [0, 1]$, define

$$y_{\tau} = (1 - \tau)x_{t+1} + \tau\epsilon, \quad u_{\tau} = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat{u}_{\omega} = F_{\omega}(y_{\tau}, \tau, h_t, z_t), \quad \mathcal{L}_{\text{flow}} = \|\hat{u}_{\omega} - u_{\tau}\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

Write–read cycle. The one-step endpoint $\hat{x}_0 = y_{\tau} - \tau\hat{u}_{\omega}$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0), R_{\bar{\theta}}(x_{t+1}^{(1)}), R_{\bar{\theta}}(x_{t+1}^{(2)})\right).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat{x}_{t+1} = \arg \max_{c \in \mathcal{C}_t} E_{\psi}(h_t, z_t, R_{\bar{\theta}}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

Model-induced visual rollout. V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}_{\text{energy}}^{\text{roll}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

Calibrated context and sampled identity. V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE} \left(R_{\bar{\theta}}(\hat{x}_0^{K=2}), \{R_{\bar{\theta}}(y^{(v)})\}_{v=1}^V \right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

4 Predictive Visual Field

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R(x_{t+1}) \in \mathbb{S}^{d-1}$, a deterministic proposal predicts

$$\mu_t = \frac{P_\psi([h_t, z_t])}{\|P_\psi([h_t, z_t])\|_2}.$$

The proposal is a continuous image-derived state, not a class index. It is trained with geodesic alignment, image contrast, and full-history advantage against a detached last-image branch.

In parallel, choose random unit source ϵ , let q_τ follow the shortest spherical geodesic from z_{t+1} to ϵ , and let u_τ be its tangent velocity. The conditional field learns

$$\mathcal{L}_{\text{state-flow}} = \mathbb{E} \|v_\eta(q_\tau, \tau, [h_t, z_t]) - u_\tau\|_2^2.$$

The loss is full tangent-vector energy, not coordinate-mean MSE. Sampling integrates backward with the spherical exponential map,

$$q_{k-1} = \text{Exp}_{q_k}(-\Delta\tau v_\eta(q_k, \tau_k, [h_t, z_t])),$$

Predictive Visual Field

Language evolves as a distribution over continuous retinal states; a separate visual actuator writes each state as ink.

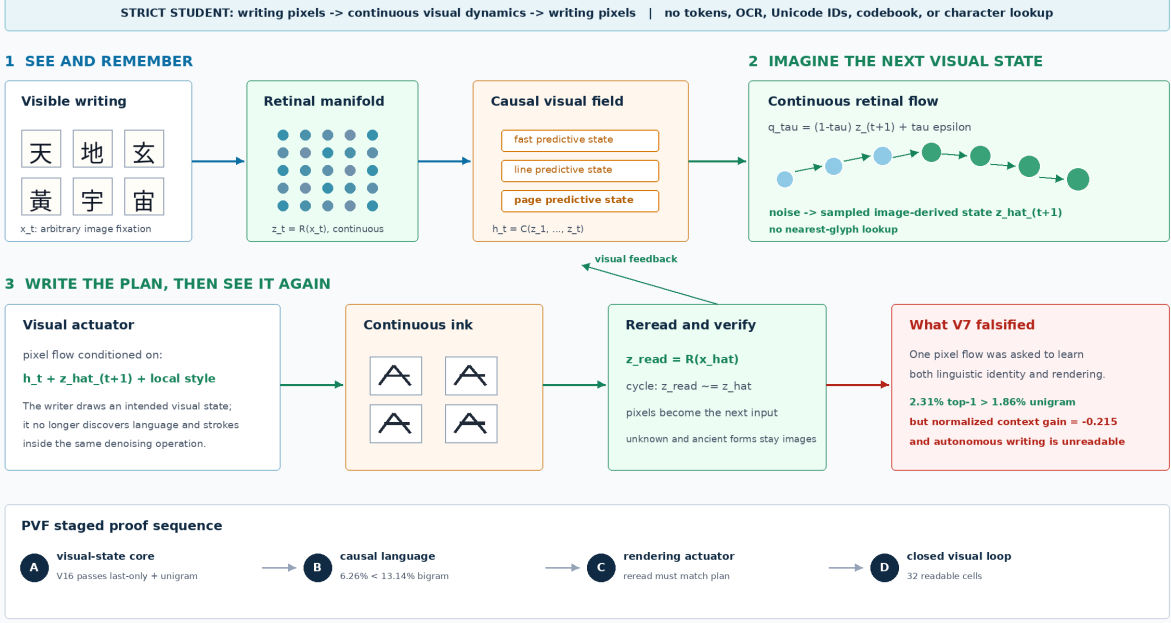


Figure 3: Predictive Visual Field factorization. A causal image field predicts a low-variance continuous visual proposal and a stochastic distribution over image-derived retinal states. A later separate actuator renders a selected state and the retina rereads its pixels.

yielding intended next visual states $\hat{z}_{t+1}$, not glyph IDs. Candidate images are used only by the evaluator: it rereads them with the frozen retina and computes kernel density against continuous samples. There is no learned 512-way output layer.

The training-only multiview anchor curriculum contains image tensors. Positives are inferred from retinal cosine similarity; labels and target indices do not enter the student, and the bank is not deployed. This visual contrast is necessary because geodesic endpoint alignment alone can reward a visual mean without separating language alternatives.

V16 augments the pretrained recurrent trajectory $h_{1:t}^R$ with three residual causal memory blocks. Each block combines a left-padded depthwise temporal field at dilation 1, 2, or 4, global causal attention, and a gated feed-forward residual. For memory M_ϕ , the corrected state is

$$h_t^{V16} = h_t^R + \sigma(g)W_o \text{LN}\{M_\phi(h_{1:t}^R, z_{1:t})\}.$$

The per-channel gate is initialized at $g = -4$, preserving the trained V15 function at the start of the upgrade. The selected checkpoint has mean $\sigma(g) = 0.018023$; therefore the frozen result is a bounded memory pilot, not evidence that a large residual correction is necessary.

V17 implements a separate visual actuator that writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda [1 - \cos\{R_{\hat{\theta}}(\hat{x}_{t+1}), z_{t+1}\}].$$

PVF V16: Causal Visual Memory on One RTX 4090

A 16.47M-parameter student reads writing pixels, forms continuous visual states, and predicts the next state without a symbolic vocabulary.

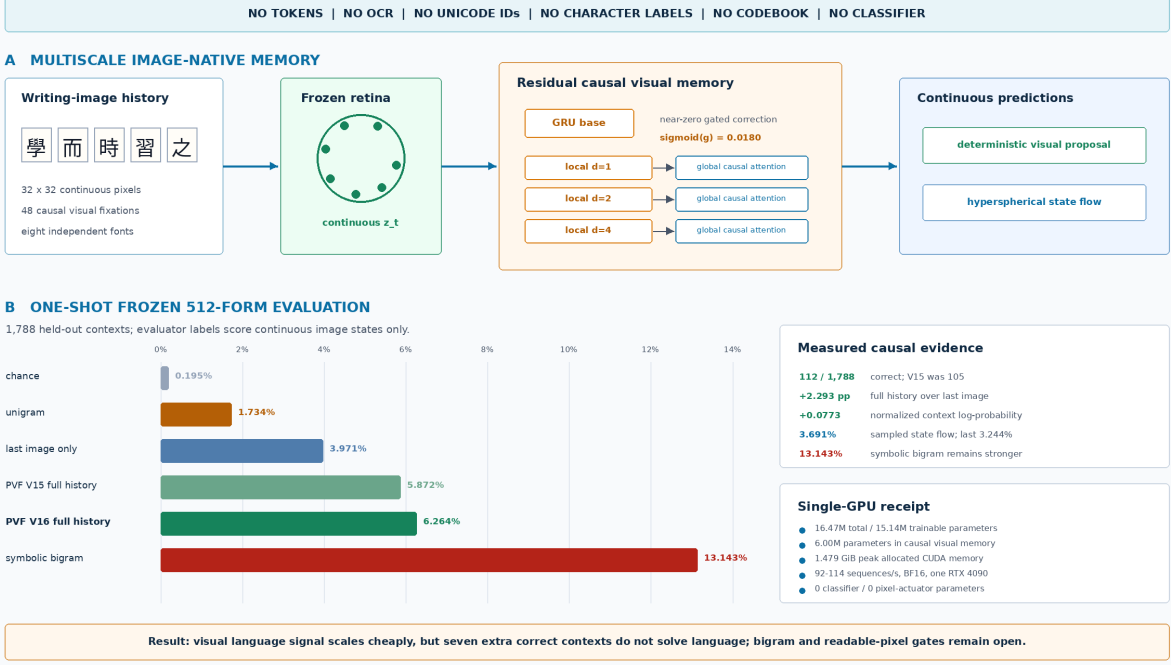


Figure 4: Measured PVF V16 memory result. Both deterministic proposal and stochastic hyperspherical field use full image history and beat random, last-only, and unigram controls. V16 improves proposal top-1 by seven frozen contexts over V15, but remains below the symbolic bigram and has no pixel actuator.

No nearest-glyph projection, character classifier, or token unembedding is permitted. In V17, $\hat{z}_{t+1}$ is supplied by a different-font image of the target form so actuation is isolated from language prediction. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [31]. It combines the next-fixation representation objective of recurrent JEPA [29] with the prediction/generation separation explored by D-JEPA [30], and uses intrinsic flow matching on a Riemannian manifold [26, 27], while retaining an image-derived language state.

5 Isolated Visual Actuation and Motor Planning

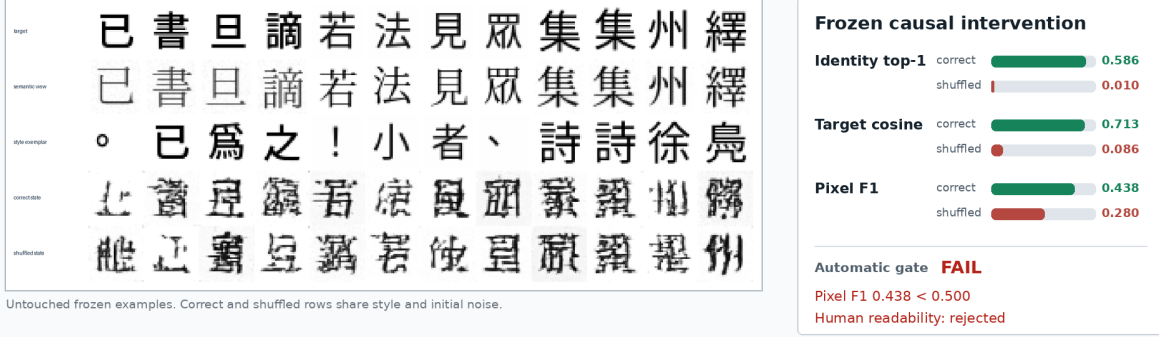
The V17 actuator tests whether an image-derived continuous state can control new pixels without a glyph table. A frozen retina produces $z = R(x^{\text{semantic}}) \in \mathbb{S}^{191}$ from one font view, and a convolutional encoder produces style vector $s = E_{\omega}(x^{\text{style}})$ from a different character in the desired target render. A 5.73M-parameter conditioned U-Net integrates rectified flow from Gaussian noise to a 32×32 ink image. The target image enters losses only; the spatial condition to the writer is a constant blank plane.

Training combines stroke-weighted flow matching, an endpoint loss, retinal cosine, multi-positive retinal contrast, and gradients through a two-step deployed sampler. At evaluation, correct and shuffled branches receive identical style images and initial noise. Only the intended state is rolled

V17: continuous visual state controls pixels, not yet readable strokes

One RTX 4090 | 5.73M trainable actuator parameters | no token IDs, OCR, Unicode IDs, labels, or glyph lookup

different-font image -> frozen retina -> continuous intended state + style image -> pixel flow -> generated ink -> reread



Measured verdict visual identity is causally controlled; exact stroke topology and readability are not solved

Next: decode the continuous state into a supervised spatial motor plan, then use stochastic flow only for refinement

Figure 5: Measured V17 frozen result. Continuous intended state strongly controls recovered visual identity under a same-noise shuffled-state intervention, but pixel F1 and human readability fail. The correct-state row contains state-dependent pseudo-characters, not reliably readable target forms.

within the batch. The global evaluator retrieves among all 512 generated examples, with duplicate positives inferred from image-state similarity rather than labels.

V17 was selected at step 1,600 by a preregistered development rule and queried the frozen record split once. Correct-state top-1 is 58.594%, versus 0.977% shuffled; target cosine is 0.71301 versus 0.08612; and style-copy cosine is only 0.05858. These measurements reject style copying as the explanation for the causal signal. Pixel F1 is 0.43849 versus 0.28001 shuffled, failing the required 0.50, and human review rejects most outputs as unreadable. Training takes 339.67 seconds and peaks at 1.588 GiB allocated CUDA memory on one RTX 4090.

The result identifies a topology gap rather than an identity-control gap.

5.1 Topology-first visual motor plan V18

V18 decodes state and style into a directly supervised spatial visual motor plan,

$$p_\omega = D_\omega(z, E_s(x^{\text{style}})) \in [0, 1]^{H \times W}.$$

A context MLP modulates a learned 4×4 seed, conditioned residual blocks upsample it to 32×32 , and a sigmoid emits continuous ink. The loss joins stroke-weighted binary cross entropy, soft Dice, pixel and Sobel-edge distances, retinal identity and contrast, and a correct-versus-shuffled state margin. Target pixels supervise the output but never enter the condition. The plan remains continuous and image-derived; it is not copied target ink, an ID lookup, or a discrete visual codebook. Stochastic flow is reserved for later residual style and surface variation rather than stroke placement.

V18: a 2.36M visual motor plan learns readable writing topology

One RTX 4090 | continuous image-derived intent | no tokens, OCR, Unicode IDs, labels, lookup, or codebook

different-font image -> frozen retina -> continuous intent + style image -> deterministic spatial motor plan -> ink



Fresh development renderings. Correct and shuffled rows share style; only continuous intent changes.

Measured consequence

structured writing is learnable as continuous image topology with a small consumer-GPU model
autonomous language choice, complex-form fidelity, and closed-loop readability remain unsolved

DEVELOPMENT-ONLY EVIDENCE | selected step 1,400 | 1,600 updates | 458.34 s | 0.778 GiB peak allocated CUDA

Figure 6: Measured V18 development result. A 2.36M-parameter deterministic motor planner turns continuous image-derived intent into recognizable writing. Shuffling only intended state destroys identity and topology while preserving mean ink occupancy. This is development-only evidence; the frozen bank remains untouched.

The salted partition contains 6,573 training, 223 development, and 221 frozen records. Selection maximized development correct-state pixel F1 subject to fixed topology, state-dependence, retinal-similarity, and non-copying gates. Step 1,400 was selected at pixel F1 0.6890 versus 0.3368 shuffled and global identity top-1 78.32% versus 1.17%. A fresh development seed yields 512 examples: identity top-1 is 73.63% versus 0.98%, target cosine is 0.8462 versus 0.0716, and pixel F1 is 0.6577 versus 0.3129. Mean ink fraction is 0.17505 in both branches and style-copy cosine is 0.0686.

Full-resolution review finds clear simple and medium forms but merged strokes in dense forms. Because the preregistration required human readability without defining a numeric blinded threshold, post-hoc inspection cannot authorize a frozen query. V18 therefore accepts readable topology emergence on development, rejects broad-complexity readability, and leaves the frozen identifier bank sealed. The next experiment must define complexity strata and a recognition rubric prospectively.

5.2 Spatial-retinal residual V19 and its rejection

V18 compresses the semantic image into one global vector before drawing. V19 tests the narrower hypothesis that this pooling discards component location in dense forms. The frozen retina now exposes both its existing global state and its pre-pooling spatial field,

$$(z, F) = R(x^{\text{semantic}}), \quad z \in \mathbb{S}^{191}, \quad F \in \mathbb{R}^{192 \times 4 \times 4}.$$

V19 development gate	Measured	Required	Result
Overall pixel F1	0.6710	> 0.68	fail
Dense pixel F1	0.7278	> 0.58	pass
Dense gain over shuffled field	0.0088	> 0.12	fail
Dense gain over zero field	0.0054	> 0.03	fail
Identity top-1	72.66%	> 75%	fail
Target cosine	0.8416	> 0.84	pass

Table 1: Fresh 512-candidate V19 development audit. Four fixed automatic gates fail, so the blinded human review and frozen evaluation are not authorized.

Let $c = [z, E_s(x^{\text{style}})]$ and let $q_0 = S(c)$ be the learned global 4×4 seed of a V18-architecture planner. A conditioned convolutional adapter supplies a spatial residual,

$$a = A_\varphi(F, c), \quad q = q_0 + \sigma(g)a, \quad p = D(q, c).$$

The adapter’s final projection is initialized exactly to zero, so V19 initially equals the selected global planner for every input even though its scalar gate is trainable. All global-planner and retina parameters remain frozen. This makes a gain attributable to spatial information rather than unrestricted baseline fine-tuning.

A clean causal comparison requires a new global baseline trained from scratch on the V19 partition. Loading V18 would contaminate V19 because V18’s training split can intersect the new holdout. The selected clean global checkpoint is therefore frozen before the spatial adapter is trained; no V17 or V18 learned weight enters the experiment.

Complexity is computed from target image geometry, never a character label. For thresholded ink b , define

$$C(b) = \bar{b} + \frac{1}{2} \left(|\overline{\Delta_x b}| + |\overline{\Delta_y b}| \right) + 0.1 \mathbf{1}[\text{AvgPool}_4(b) > 0.05].$$

The preregistered strata are simple $C < 0.24$, medium $0.24 \leq C < 0.35$, and dense $C \geq 0.35$, weighted 1, 1.25, and 2 during spatial training. The objective is

$$\begin{aligned} \mathcal{L}_{19} = & \mathcal{L}_{\text{WBCE}} + \mathcal{L}_{\text{Dice}} + 0.5\mathcal{L}_1 + 0.25\mathcal{L}_{\text{edge}} \\ & + 0.05\mathcal{L}_{\text{id}} + 0.05\mathcal{L}_{\text{NCE}} + 0.1\mathcal{L}_{\text{spatial-margin}} + 0.1\mathcal{L}_{\text{zero-margin}}. \end{aligned}$$

Rowwise terms use the complexity weights. The margins require correct fields to outperform batch-shuffled fields by 0.03 and the zero-field branch by 0.01 in target pixel error.

Evaluation fixes target and style while comparing five branches: correct global state plus correct field; correct global plus shuffled field; shuffled global plus correct field; both shuffled; and correct global plus zero field. The prospective gate requires overall pixel F1 above 0.68, dense F1 above 0.58, dense spatial-shuffle and zero-field advantages, global identity above 0.75, and 38/48 blinded recognitions with stratum-specific minima. The implementation passes exact-equivalence, gradient-isolation, field-shape, intervention, complexity, and serialization tests.

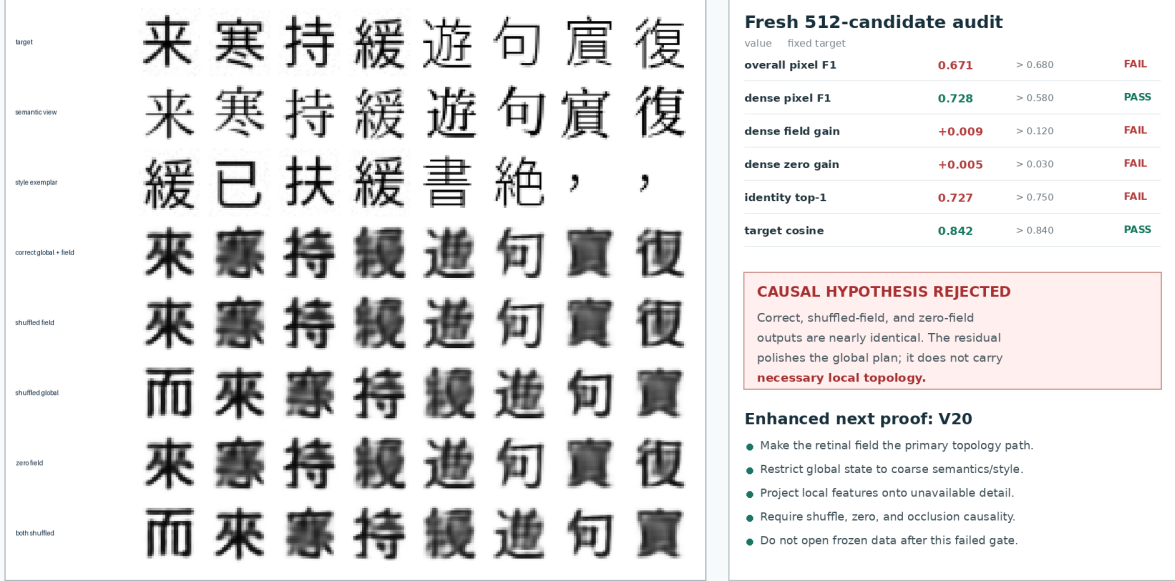
A clean global planner was trained from scratch for 1,600 updates on the V19 partition, selected at step 1,600, and frozen. It reached development F1 0.6754, identity top-1 77.93%, and target cosine 0.8339. The V19 intervention then trained only 764,545 adapter parameters for 1,600 updates in 329.51 seconds, using 0.899 GiB peak allocated CUDA memory on one RTX 4090. A separate deterministic audit rendered 512 new development candidates. Table 1 records the fixed decision.

This result rejects the additive routing mechanism, not the possibility that the field contains useful information. The global planner can already draw the whole form, so ordinary reconstruc-

V19: spatial retinal residual fails its causal topology test

One RTX 4090 | 764,545 trainable parameters | fixed protocol | frozen split sealed

global visual plan + gated 4x4 retinal-field residual -> ink | correct / shuffled / zero-field interventions



DEVELOPMENT-ONLY NEGATIVE RESULT | 1,600 updates | 329.51 s | 0.899 GiB peak CUDA | human audit not authorized

Measured consequence: adding spatial features is insufficient when a complete global writer can ignore them.

Figure 7: Measured V19 result. Correct, shuffled-field, and zero-field rows are nearly identical, while shuffling the global state changes identity. The spatial adapter learned a shared residual polish rather than necessary local topology.

tion gradients need not use the field. The next bounded candidate must make local visual state structurally necessary.

5.3 Retinal topology router V20: local causality without writer selection

V20 implements the preregistered coarse/detail decomposition

$$p = \sigma(U(G(z, s)) + (I - UD)H(F, s)),$$

where G emits only one logit per 8×8 output block and H maps each 4×4 retinal-field cell pointwise to its corresponding 8×8 detail patch. The fixed projection $I - UD$ removes each patch mean, so the global route cannot represent within-block detail. An exactly capacity-matched control replaces F with $\text{tile}(z)$ while retaining the same local decoder, losses, and operations. Each arm has 506,448 trainable parameters.

The new salted split contains 6,611 training, 196 development, and 210 frozen records. Both arms train for 1,600 updates against the same frozen V16 retina. The fixed interventions hold target and style constant while shuffling the local field, zeroing it, shuffling global state, shuffling both, or occluding each field quadrant. Validation renders 512 development candidates, including 146 dense forms. Table 2 records the final endpoints; neither is a selected checkpoint.

The candidate’s correct dense F1 exceeds shuffled-field by 0.12177 and zero-field by 0.36128; its quadrant locality is exactly 1.0. These measurements establish that the continuous field carries

V20 development gate	Candidate	Required	Result
Overall pixel F1	0.6361	> 0.66	fail
Dense pixel F1	0.7013	> 0.68	pass
Dense gain over shuffled field	0.1218	> 0.12	pass
Dense gain over zero field	0.3613	> 0.10	pass
Identity top-1	75.00%	> 70%	pass
Target cosine	0.8152	> 0.82	fail
Occlusion locality	1.0000	> 0.40	pass
Detail block-mean magnitude	2.03×10^{-6}	< 10^{-6}	fail
Dense gain over matched control	0.0179	> 0.03	fail

Table 2: Final V20 development endpoints. The field candidate passes every local-causality gate but fails writer selection and the matched-control margin. Because neither arm selects, the formal paired audit, blinded review, and frozen evaluation remain forbidden.

necessary, topographically aligned detail. They do not establish a selected writer. Overall F1 and reread cosine remain below threshold. Both arms also exceed the strict 10^{-6} detail-mean tolerance after detail magnitude grows, indicating that floating-point recentering is a poor way to encode an algebraic invariant. Candidate dense F1 exceeds the control by only 0.01791 at the final endpoint, below the paired 0.03 margin. Candidate and control train in 325.90 and 327.11 seconds and peak at 0.323 and 0.397 GiB allocated CUDA memory, respectively.

5.4 Field-complete writer V21: complete local route, rejected rasterizer

V21 tests the field-complete correction rather than widening V20. Global state and image-derived style emit only spatially uniform modulation. Each local field cell independently emits one coarse coefficient and 63 detail coefficients,

$$(\gamma, \beta) = M(z, s), \quad (c_{ij}, a_{ij}) = A_F(\gamma \odot F_{ij} + \beta), \quad \ell_{ij} = c_{ij}\mathbf{1} + B_0 a_{ij},$$

where B_0 contains the 63 non-DC columns of a normalized Walsh–Hadamard basis. Its column sums and Gram error are exactly zero in the recorded FP32 computation. The 16 nonoverlapping 8×8 patches make the 32×32 logit field. There is no coordinate input, position parameter, cell mixing, global spatial projection, or learned glyph table. The matched control tiles z into all 16 source cells while retaining every parameter and operation. Each arm has 582,336 trainable parameters.

The preregistered salted split has 6,613 training, 195 development, and 209 frozen records. Both arms train for exactly 1,600 updates. No candidate checkpoint passes every selection gate. Table 3 therefore reports step 1,400 only as the best diagnostic snapshot under the fixed ranking, not as a selected model.

The tiled-global control selects its structural step 1,200 and reaches only 0.1473 overall F1, 0.3074 dense F1, 25.00% identity top-1, and 0.4332 target cosine. Candidate step 1,400 exceeds those values by 0.4564 overall F1 and 0.3979 dense F1. This comparison is descriptive rather than the formal paired audit: the snapshots use different development draws and steps, and the paired evaluator requires a selected candidate before constructing a shared audit. It must therefore refuse V21.

Candidate and control train in 327.04 and 333.41 seconds at 0.325 and 0.400 GiB peak allocated CUDA memory. Frozen-image count remains zero. V21 therefore establishes complete, necessary, topographic use of continuous local state, but rejects nonoverlapping pointwise patch emission as

V20: local retinal topology becomes necessary, writer gate still fails

One RTX 4090 | 506,448 parameters per arm | fixed development protocol | frozen split sealed

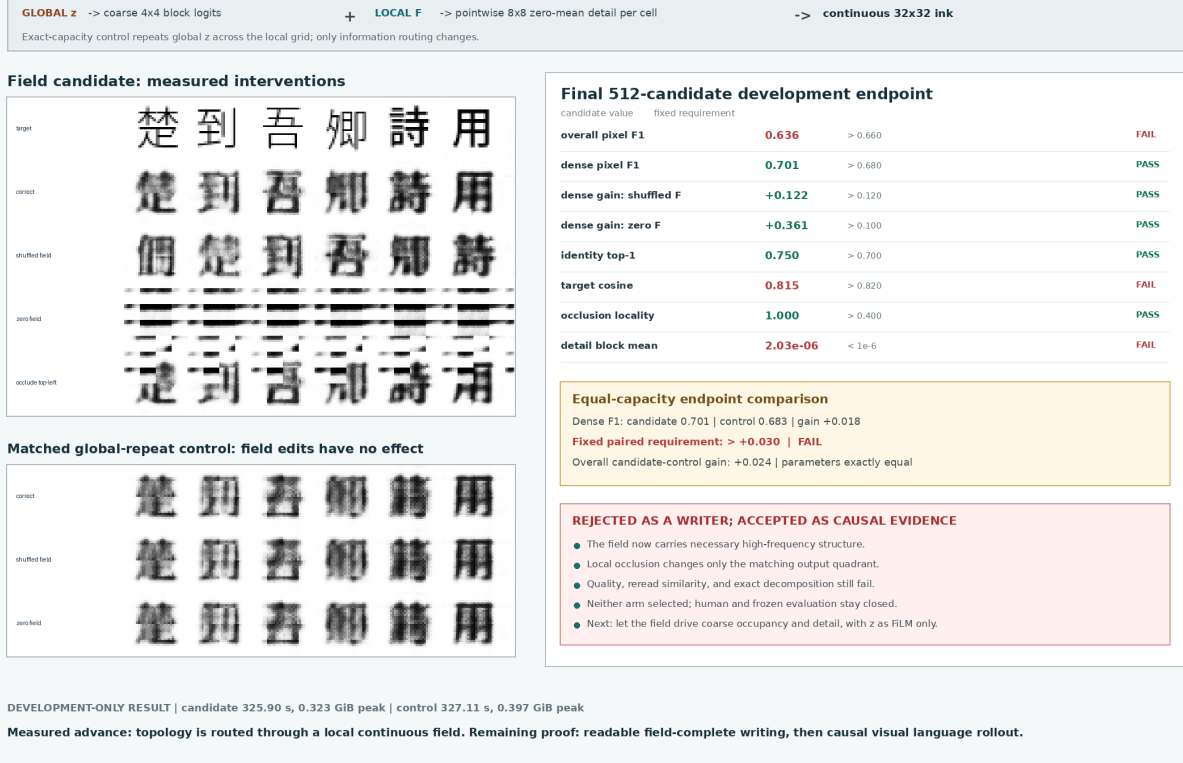


Figure 8: Measured V20 result. Shuffling or zeroing the candidate’s local field destroys detail, and a quadrant occlusion changes only its corresponding output quadrant. The equal-parameter global-repeat control is invariant to field interventions. Recognizable endpoints do not pass the fixed writer and paired-control gates.

a broadly readable writer. The next writer test should add overlapping local support with a fixed partition-of-unity blend, or an equivalently bounded multiscale local operator, while retaining an exact-parameter tiled-global control and a measured influence radius.

5.5 Visual Binding Stream V22: image prompt, rejected relation

V22 removes supplied semantic intent and asks the smallest prompt-following question that also requires arbitrary visible form. Every episode is a six-frame image stream,

$$X = (l_1, g_1, l_2, g_2, o, q),$$

where two visible labels bind two distinct, previously unseen glyph images. The operation image is either the visible form for same or other, and the final query is one of the two label images. A paired counterfactual changes only q ; the correct canonical answer image must switch. Source glyphs use independent noncanonical faces, so exact pixel copying cannot satisfy the target.

The frozen V16 retina maps each frame to global and local continuous states. A four-block visual Transformer forms a query from its final hidden state and a softmax over all six frames. Weighted global/local values feed an overlapping 12×12 , stride-8 local writer with a fixed positive partition-of-unity blend. The query-blind control replaces only the final query retinal state with a learned

V21 candidate development gate	Measured	Required	Result
Overall pixel F1	0.6038	> 0.66	fail
Simple pixel F1	0.5648	> 0.58	fail
Medium pixel F1	0.5945	> 0.60	fail
Dense pixel F1	0.7053	> 0.70	pass
Dense gain over shuffled field	0.1739	> 0.15	pass
Dense gain over zero field	0.3465	> 0.20	pass
Identity top-1	79.10%	> 74%	pass
Target cosine	0.8331	> 0.82	pass
Occlusion pixel change	0.0926	> 0.03	pass
Occlusion locality	1.0000	> 0.95	pass
Detail block-mean magnitude	3.87×10^{-7}	$< 5 \times 10^{-6}$	pass
Decomposition error	0.0	$< 10^{-6}$	pass

Table 3: Best V21 candidate diagnostic snapshot. It passes every causal, identity, reread, and algebraic gate but fails overall, simple, and medium pixel fidelity. It is not a selected checkpoint.

V22 candidate development gate	Measured	Required	Result
Binary choice	0.4824	> 0.85	fail
Counterfactual switch	0.0078	> 0.80	fail
Held-out-combination switch	0.0113	> 0.75	fail
Identity top-1	0.1592	> 0.45	fail
Identity gain over query shuffle	0.0225	> 0.20	fail
Target cosine	0.4601	> 0.78	fail
Pixel F1	0.5107	> 0.58	fail
Oracle-writer F1	0.6020	> 0.64	fail
Paired output L1	0.0089	> 0.08	fail
Frozen images instantiated	0	0	pass

Table 4: V22 query-aware endpoint. The anti-copy margins and structural receipts pass, but every prompt-dependence and output-quality gate fails. The query-blind control ends at 0 switch accuracy, 0.1533 identity top-1, 0.5125 pixel F1, and exactly zero paired-output difference.

continuous null state. Both arms contain exactly 3,410,128 trainable parameters with identical shapes. The student path contains no string, token or Unicode ID, OCR, character/operation label, answer index, glyph lookup, finite codebook, or external language model.

The fixed identity bank contains 815 training, 104 development, and 105 frozen Chinese forms under a salted split; two operation/label combinations are held out compositionally. Each arm trains for exactly 1,600 updates. Table 4 reports the final development endpoints. Candidate and control writer/identity trajectories nearly overlap, while prompt switching remains absent.

The mechanism audit makes the rejection more specific than failure to converge. Candidate mean attention is 1.0 on the operation frame, 2.49×10^{-12} and 5.53×10^{-12} on the two glyph frames, and 1.37×10^{-13} on the query. The entropy penalty rewards this confident shortcut. The task, however, is a relation among operation, query, labels, and glyphs; no single frame contains the answer. Candidate training takes 258.70 seconds at 0.315 GiB peak allocated CUDA memory, so model/GPU capacity is not the identified bottleneck.

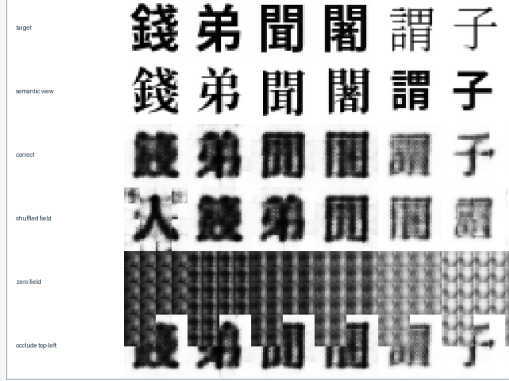
No candidate checkpoint selects. The paired evaluator refuses before fresh data construction, blinded review is unauthorized, and no frozen image is instantiated. V22 establishes a real image-prompt to image-answer interface and weak unseen-form writing only; it does not establish prompt

V21: a complete local field writes Chinese structure, broad writer gate still fails

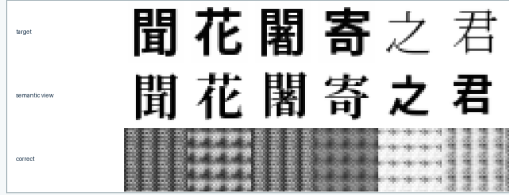
One RTX 4090 | 582,336 parameters per arm | image-only student | frozen split sealed

LOCAL F[i,j] -> shared pointwise writer -> coarse scalar + 63 zero-DC basis coefficients -> complete 8x8 patch
Global z + style provide one spatially uniform modulation. No coordinate input or cell mixing can draw around the local field.

Field-complete candidate: best diagnostic step 1400



Tiled-global control: selected structural step 1200



DEVELOPMENT ONLY | candidate 327.04 s / 0.325 GiB | control 333.41 s / 0.400 GiB | frozen images: 0

Best 512-candidate development snapshot

candidate value	fixed requirement	
overall pixel F1	0.604	> 0.660 FAIL
simple pixel F1	0.565	> 0.580 FAIL
medium pixel F1	0.594	> 0.600 FAIL
dense pixel F1	0.705	> 0.700 PASS
dense gain: shuffled F	+0.174	> 0.150 PASS
dense gain: zero F	+0.347	> 0.200 PASS
identity top-1	0.791	> 0.740 PASS
target cosine	0.833	> 0.820 PASS
occlusion locality	1.000	> 0.950 PASS
detail block mean	3.87e-07	< 5e-6 PASS

Descriptive equal-capacity comparison

Overall F1: 0.604 vs 0.147 (+0.456)
Dense F1: 0.705 vs 0.307 (+0.398)

Not a formal paired audit: the candidate did not select.

WRITER REJECTED; FIELD-COMPLETE CAUSAL ROUTE SUPPORTED

- Local state carries coarse occupancy and fine detail.
- The exact basis and every intervention invariant pass.
- The matched global-only source collapses to repeated patches.
- Simple and medium fidelity remain below fixed thresholds.
- No paired audit, human review, or frozen query is permitted.
- Next: improve local raster continuity, then predict visual answer streams.

Figure 9: Measured V21 result. The candidate uses local state for complete spatial structure: shuffling or zeroing the field corrupts the form and a local occlusion stays local. The equal-parameter tiled-global source repeats one patch. Disjoint candidate patches remain blurred and seamed, causing the preregistered broad-fidelity rejection.

understanding.

V23 removes unrestricted single-frame selection. A shared visual comparator must match query and label images, while an operation-image reader provides a continuous gate,

$$a_i = C_\theta(R(q), R(l_i)), \quad m_i = \text{softmax}_{i \in \{1,2\}}(a_i), \quad s = \sigma(U_\theta(R(o))),$$

$$w_i = sm_i + (1 - s)(1 - m_i), \quad (\hat{z}, \hat{F}) = \sum_{i=1}^2 w_i V(R(g_i)).$$

This relation circuit composes multiple visible roles before writing. Its operation semantics and equality function are learned from answer images; no symbolic target index enters the student. Exact query-blind, operation-blind, pair-swap, query-counterfactual, and operation-counterfactual interventions are required before any longer answer stream. A qualified writer should be frozen first so binder/writer co-adaptation cannot hide another average-glyph route.

5.6 Visual Relation Circuit V23: bounded prompt relation accepted

V23 fixes the V22 causal defect while preserving the same strict visual interface. Its bank contains 817 training, 109 development, and 98 frozen Chinese identities under the preregistered salt

V22: visible prompt binding fails through an operation-frame shortcut

One RTX 4090 | 3,410,128 trainable parameters per arm | image-only input/output | frozen identities sealed

Six-frame visual prompt and measured candidate selector behavior

frame 1 label 1 mean 3.4e-13 argmax 0/1024	frame 2 glyph 1 mean 2.5e-12 argmax 0/1024	frame 3 label 2 mean 1.6e-12 argmax 0/1024	frame 4 glyph 2 mean 5.5e-12 argmax 0/1024	frame 5 operation mean 100.0% argmax 1024/1024	frame 6 query label mean 1.4e-13 argmax 0/1024
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Actual endpoint samples

Only the final query image changes inside each pair; targets switch, predictions barely do.

query-aware candidate

seen	heldout
甲 田 乙 論 同 甲	地 歌 天 吾 同 地
pred target cf pred cf target	pred target cf pred cf target
田 田 論 論	歌 歌 吾 吾

query-blind control

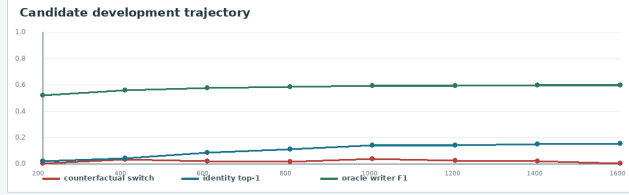
seen	heldout
甲 田 乙 論 同 甲	地 歌 天 吾 同 地
pred target cf pred cf target	pred target cf pred cf target
田 田 論 論	歌 歌 吾 吾

Fixed candidate gates at step 1,600

measured candidate	fixed requirement		
binary choice	0.482	> 0.850	FAIL
counterfactual switch	0.008	> 0.800	FAIL
held-out switch	0.011	> 0.750	FAIL
identity top-1	0.159	> 0.450	FAIL
identity gain: query	+0.022	> 0.200	FAIL
pixel F1	0.511	> 0.580	FAIL
oracle-writer F1	0.602	> 0.640	FAIL
paired output L1	0.0089	> 0.080	FAIL
target vs op. margin	+0.434	> 0.150	PASS
frozen images	0	= 0	PASS

Candidate vs query-blind control endpoint

switch 0.0078 vs 0.0000 | identity 0.1592 vs 0.1533 | F1 0.5107 vs 0.5125



DEVELOPMENT ONLY | candidate 258.70 s / 0.315 GiB | control 255.11 s / 0.316 GiB | no selected candidate

BINDING MECHANISM REJECTED

- The writer acquires weak unseen-glyph structure.
- The candidate does not bind query to either glyph.
- Entropy minimization makes the fixed operation marker
- the easiest single-frame shortcut: 1,024/1,024 argmaxes.
- No candidate checkpoint selects; paired audit refuses.
- No human review or frozen identity image is permitted.
- Next: explicit multi-frame visual relation composition.

Figure 10: Measured V22 result. Actual endpoint samples show that targets switch when only the final visual query changes, while candidate predictions barely move and match the query-blind control. The candidate selector assigns every one of 1,024 development argmaxes to the operation frame. Writer metrics improve without relational prompt binding.

visual-relation-circuit-v23. The two held-out operation/label compositions never occur during training, although every individual operation and label pair does. Every model call receives only floating image tensors or continuous states derived from those tensors; renderer strings and episode metadata are deleted before **forward**.

Stage A independently trains a 1,122,081-parameter convolutional U-Net to map a noncanonical source-glyph image to a canonical target image. It has no identity embedding, target-font ID, coordinate channel, or character table. The selected step-1,000 checkpoint reaches 0.7583 pixel F1, 0.9951 identity top-1, and 0.9375 target cosine on all 109 unseen development identities. Its pixel-F1 gains over the raw source and a shuffled source are 0.2173 and 0.4237. The canonicalizer is then frozen before relation training.

Stage B uses normalized frozen-retina states. For each label image,

$$c_i = \langle R(q), R(l_i) \rangle, \quad m_i = \text{softmax}_{i \in \{1,2\}}(\tau c_i),$$

where the positive temperature τ is learned. A small operation-image reader produces

$$s = \sigma(U_\theta(R(o))),$$

Fresh paired metric	Relation-aware	Query-blind	Operation-blind
Binary choice	0.9990	0.5010	0.5002
Query switch	0.9980	0.0000	0.0615
Operation switch	0.9980	0.0156	0.0000
Identity top-1	0.9951	0.2947	0.2993
Pixel F1	0.7563	0.5407	0.5404
Query-output L1	0.1915	0.0000	0.0109
Operation-output L1	0.1912	0.0017	0.0000
Pair-swap output L1	0.0000	0.0000	0.0000

Table 5: V23 fresh paired development audit. A control has exactly zero response to its blinded factor by construction, while the relation-aware arm retains both dependencies with matched parameter shapes.

and composes match and operation before generation:

$$w_i = sm_i + (1 - s)(1 - m_i), \quad x_r = w_1g_1 + w_2g_2, \quad \hat{y} = C_\psi(x_r).$$

Thus same routes the visually matched source and other routes its complement. The operation semantics are learned from answer-image loss, not supplied as a class label. The routed variable is a continuous source-pixel field, and C_ψ is the frozen Stage A canonicalizer. The trainable relation circuit has 25,602 parameters: the operation reader, one temperature scalar, and two continuous null states.

Candidate, query-blind, and operation-blind arms have exactly identical parameter names and shapes. Query-blind replaces only $R(q)$ with its learned continuous null state; operation-blind replaces only $R(o)$. All arms process base, query-counterfactual, operation-counterfactual, and pair-swapped image prompts jointly for 600 fixed updates. The selected candidate reaches 0.9946 binary choice, 0.9902 query switch, 0.9902 operation switch, 0.9873 identity top-1, and 0.7548 pixel F1 on checkpoint-selection development data.

A fresh paired audit then constructs 1,024 new development episodes and 4,096 prompt variants. Table 5 reports the exact interventions. The candidate changes identity and pixels when either visible causal factor changes. Each blind control is exactly invariant to the factor it cannot see, and all arms remain exactly invariant to pair order.

After the paired gate passes, an opaque development review presents 48 image cards without transcriptions, targets, correctness labels, or frozen examples. The reviewing agent chooses which of the two visible source forms the generated answer depicts before the scorer opens the sealed key. It obtains 48/48 overall and 12/12 on held-out compositions, above the fixed 44/48 and 11/12 gates. This is an agent visual review, not a human-subject study.

Only then does the frozen evaluator instantiate the 98 held-out identities. It runs once on 1,024 episodes, 4,096 prompt variants, and four identity-bank views, with no model selection or threshold changes. Every gate in Table 6 passes. The run takes 4.74 seconds and peaks at 0.498 GiB allocated CUDA memory. Stage A training takes 214.10 seconds at 0.700 GiB; candidate relation training takes 117.18 seconds at 1.406 GiB, all on one RTX 4090.

V23 therefore accepts a bounded image-only relation: multiple visible roles are composed, unseen source forms are routed, and the answer is emitted as pixels. It does not establish arbitrary syntax or free-form generation. Frame roles and the two-pair same/other algebra remain architecturally fixed, and the answer is a canonicalized version of one visible source form. V24 implements the preregistered next test: remove absolute frame roles, consume a variable-length visual instruc-

V23 frozen gate	Measured	Required	Result
Binary choice	0.9983	> 0.95	pass
Query switch	0.9961	> 0.90	pass
Operation switch	0.9971	> 0.90	pass
Held-out minimum switch	0.9961	> 0.85	pass
Identity top-1	0.9946	> 0.75	pass
Pixel F1	0.7848	> 0.68	pass
Target cosine	0.9399	> 0.82	pass
Query-label visual match	0.9995	> 0.98	pass
Operation-gate accuracy	1.0000	> 0.98	pass
Pair-swap consistency	1.0000	> 0.99	pass

Table 6: Single authorized V23 frozen evaluation over previously unseen identities. Every threshold was fixed before implementation and remained unchanged.

tion, emit two answer-image frames, and make frame 2 causally depend on rereading the generated pixels of frame 1.

5.7 Visual Packet Reread Stream V24: causal two-frame stream accepted

V24 replaces V23’s six absolute roles with a variable sequence of three-frame visual packets. Its prompt and autonomous answer have shapes

$$X \in [0, 1]^{B \times T \times 1 \times 32 \times 32}, \quad T \in \{15, 18, 21, 24\}, \quad \hat{Y} \in [0, 1]^{B \times 2 \times 1 \times 32 \times 32}.$$

Each packet is *header*, *content A*, *content B*. The visible header glyphs for pair (U+5951), operation (U+6CD5), query (U+95EE), distractor (U+65C1), and stop (U+6B62) denote their respective roles. Packets before the terminating packet are shuffled. Training uses 15, 18, and 21 active frames; 24 frames with three distractors are held out by length. Dynamic batching adds only all-zero packet images and supplies neither active lengths nor padding masks.

The identity bank contains 829 training, 88 development, and 107 frozen Chinese forms under the fixed salt **visual-packet-reread-stream-v24**. Two operation/label-pair compositions are held out while every constituent operation and label pair appears separately in training. As in V23, renderer strings and episode metadata are deleted before every model call.

For packet j , the frozen V16 retina yields normalized header and content states H_j, A_j, B_j . Three learned normalized prototypes locate roles from header pixels:

$$e_{jr} = \tau_r \langle H_j, p_r \rangle, \quad r \in \{\text{pair}, \text{operation}, \text{query}\}.$$

Straight-through hard routing selects the top two pair packets and top one operation and query packet; its backward surrogate is a softmax. Scores inspect only header images, not packet contents. With located query q , operation o , pair labels A_j , and pair-glyph images B_j^{image} , frame 1 uses the frozen V23 comparison temperature, operation reader, and canonicalizer:

$$m_j = \text{softmax}_{j \in \text{pair}} (\tau_{23} \langle q, A_j \rangle), \quad s = \sigma(U_{23}(o)),$$

$$w_j = sm_j + (1 - s)(1 - m_j), \quad x_1 = \sum_{j \in \text{pair}} w_j B_j^{\text{image}}, \quad \hat{y}_1 = C_{23}(x_1).$$

The second step rereads the actual generated pixels:

$$r_1 = R(\hat{y}_1), \quad b_j = \text{softmax}_{j \in \text{pair}} (\tau_{23} \langle r_1, B_j \rangle),$$

V23: an image-only relation circuit follows a visible prompt

One RTX 4090 | six raster frames in, one raster answer out | 98 unseen frozen identities | no token or Unicode path

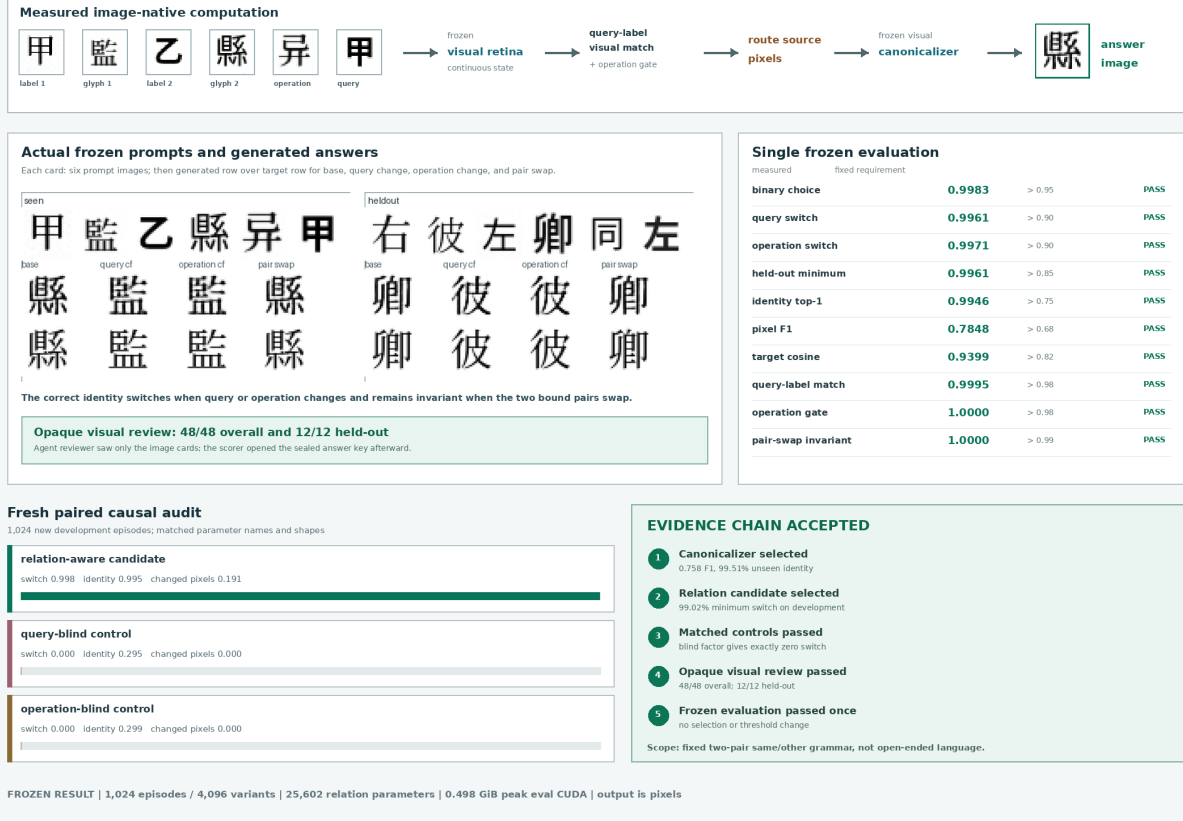


Figure 11: Measured V23 result. Actual frozen samples show generated answers above their targets for base, query-counterfactual, operation-counterfactual, and pair-swapped prompts. The matched blind controls localize both causal dependencies, and the single frozen run passes every fixed gate.

$$x_2 = \sum_{j \in \text{pair}} b_j A_j^{\text{image}}, \quad \hat{y}_2 = C_{23}(x_2).$$

Thus no pre-render identity state crosses the frame-1/frame-2 boundary. All weighted reductions accumulate in FP64.

V16 retina parameters and every inherited V23 parameter are frozen. V24 trains only three 192-dimensional role prototypes, four 192-dimensional continuous null states, and three role temperatures: 1,347 parameters total. The packet-aware, header-blind, query-blind, operation-blind, and history-blind arms have identical parameter names, shapes, and counts. Each receives exactly 800 AdamW updates in BF16 with batch 64 and eight workers on one RTX 4090. The candidate selects step 100, then completes the fixed run in 394.18 seconds at 2.322 GiB peak allocated CUDA memory.

A fresh audit uses 1,024 newly rendered development episodes at seed 22260832. Table 7 shows the paired causes. A blind arm is exactly invariant to the intervention it cannot observe. Header blindness prevents reliable role localization, while history blindness leaves frame 1 intact but breaks frame 2.

After every paired gate passes, an opaque agent visual audit presents 48 development cards containing only the packet images, two visible source choices, and two generated frames. It exposes

Fresh paired metric	Packet-aware	Corresponding blind
Query switch	0.9922	0.0000
Query-output L1	0.1677	0.0000
Operation switch	0.9932	0.0000
Operation-output L1	0.1676	0.0000
Frame-2 history switch	0.9961	0.0000
History-output L1	0.1421	0.0000
Minimum role localization	1.0000	0.0820
Frame-1 identity top-1	0.9961	0.3410 / 0.3197 / 0.2740
Frame-2 label top-1	0.9969	0.4873 / 0.3717

Table 7: V24 fresh paired development audit. The last two control cells list query/operation/header-blind and history/header-blind results, respectively. All arms have exactly 1,347 trainable parameters.

V24 frozen gate	Measured	Required	Result
Frame-1 binary choice	0.9980	> 0.95	pass
Query switch	0.9766	> 0.90	pass
Operation switch	0.9727	> 0.90	pass
Generated-history switch	0.9961	> 0.90	pass
Held-out minimum switch	0.9635	> 0.85	pass
Frame-1 identity top-1	0.9867	> 0.75	pass
Frame-2 label top-1	0.9973	> 0.95	pass
Frame-1 pixel F1	0.8371	> 0.68	pass
Frame-2 pixel F1	0.7286	> 0.58	pass
Generated-history consistency	0.9863	> 0.92	pass
Held-out $T = 24$, frame 1	0.9847	> 0.90	pass
Held-out $T = 24$, frame 2	1.0000	> 0.90	pass

Table 8: Single authorized V24 frozen evaluation. The sealed report records no model selection, no threshold change, and no repeated frozen run.

no targets, transcriptions, correctness labels, or frozen forms. The committed judgments score 47/48 for frame 1 and 48/48 for frame 2, including 12/12 for each frame at held-out $T = 24$. This is an agent visual audit, not a human-subject study.

Only then does the evaluator instantiate frozen images. It runs exactly once on 1,024 episodes, 107 unseen identities, and four identity-bank views, without model selection, threshold changes, or repetition. Table 8 reports the primary gates. All remaining role, distractor, packet-permutation, teacher-forcing, and image-boundary gates also pass. The run takes 8.13 seconds and peaks at 0.542 GiB allocated CUDA memory.

V24 therefore accepts a variable-input, two-frame image stream under one fixed visual grammar. It establishes that frame 2 can depend causally on rereading generated frame-1 pixels, not merely on a hidden state. It does not establish arbitrary sentence parsing, factual retrieval, page continuation, historical form synthesis, free-form output length, or movie generation. V25 implements the next narrower question: remove packet algebra, train on ordinary rendered Chinese, and require the resulting visual history to beat causal and frequency controls before a writer is authorized.

V24: visible packets become a causal two-frame image stream

Variable 15-24 frame prompt | generated glyph then generated label | 1,347 trainable parameters | one RTX 4090

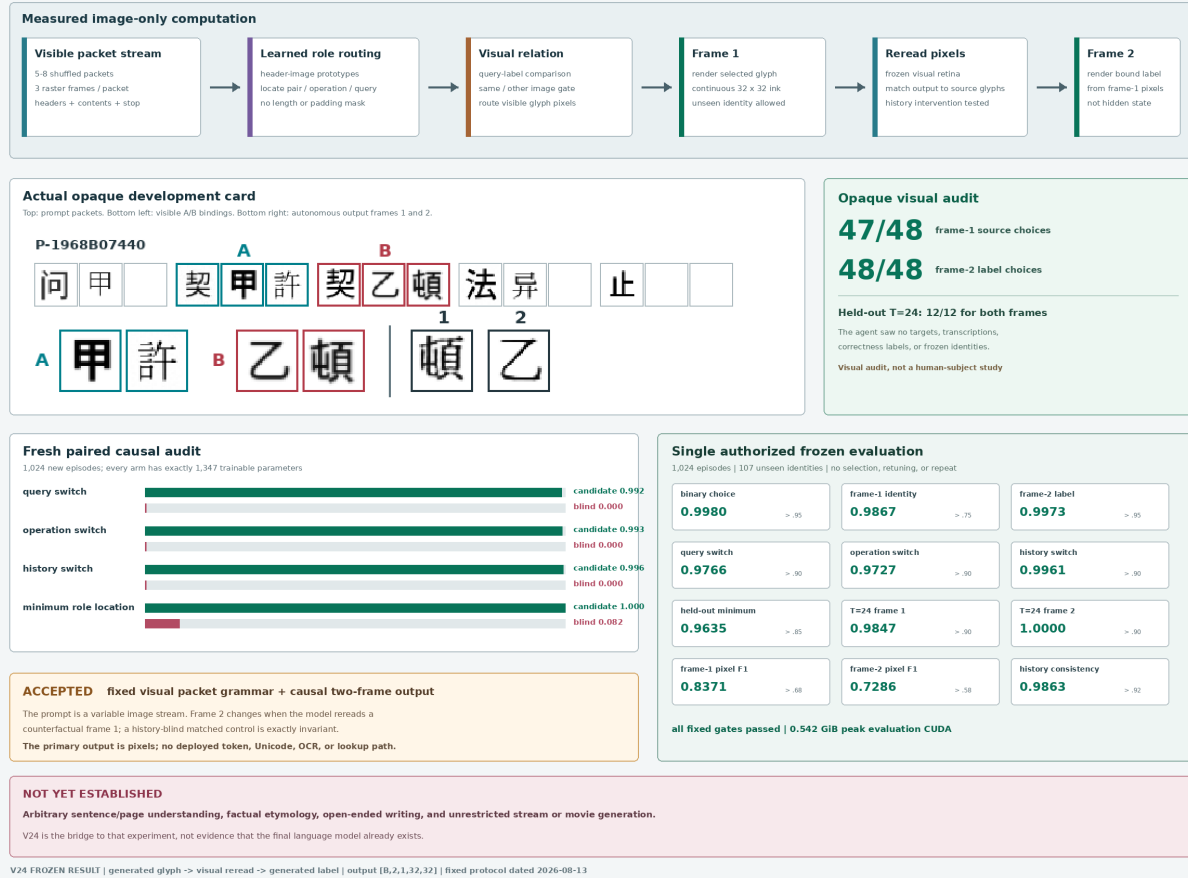


Figure 12: Measured V24 result. An actual opaque review card shows a variable-length image prompt and two autonomous answer frames. Matched controls localize header, query, operation, and generated-history causality; every gate in the single frozen evaluation passes.

5.8 Visual Cell Stream V25: natural Chinese prediction rejected

V25 treats each visible writing unit as an inspectable retinal observation and stacks observations along visual time:

$$X \in [0, 1]^{B \times T \times 1 \times 32 \times 32}, \quad T = 64.$$

The third nontrivial axis of one sample is reading order, not a character ID or geometric depth. The student receives only these grayscale images, their order, continuous corruption and flow times, random image noise, and its own generated pixels. Strings, bytes, token and Unicode IDs, OCR transcripts, character labels, candidate-bank rows, and external models are excluded from the learned path.

The fixed manifest contains 7,017 records from 16 public-domain Chinese works, with SHA-256 76048753...f6866c03. Hash partitioning by original record gives 6,608 training, 190 development, and 219 frozen groups. Training and development fonts are disjoint. Two independently rendered views of every 65-cell span reverse their context/target roles within a batch; their strings and renderer metadata are deleted before each model call. The frozen partition is counted but never rendered. An evaluator-only 1,024-form image bank and symbolic frequency tables are unavailable to the student and absent from deployment.

A frozen V16 retina maps each image to a continuous unit state $z_t \in \mathbb{S}^{191}$. Eight causal Transformer blocks of width 384 use rotary reading position and continuously corrupted prior images:

$$h_t = C_\theta(\tilde{z}_{\leq t}), \quad \hat{z}_{t+1} = \frac{P_\theta(h_t)}{\|P_\theta(h_t)\|_2}.$$

There is no character embedding or vocabulary projection. The fixed language objective combines independent-view cosine prediction with multi-positive in-batch visual contrast,

$$\mathcal{L}_{\text{language}} = 1 - \langle \hat{z}_{t+1}, z'_{t+1} \rangle + 0.5 \mathcal{L}_{\text{visual contrast}}.$$

The complete model contains 25,549,714 parameters; 15,810,241 are trainable in the language stage. The run performs 2,400 BF16 AdamW updates with physical batch three, two view directions, and four-step accumulation, for 24 effective streams per update on one RTX 4090.

Development evaluation uses 2,048 fixed windows. The same model is queried with full history, only the last cell, shuffled prior history with the last cell fixed, and blank history. The evaluator separately computes image-unigram and symbolic-bigram controls from training frequencies. It also forms 512 pairs with the same last cell but different histories and different true next cells. Table 9 reports every selection-relevant measure.

Thus the causal field uses ordered history, but does not use it accurately enough to select the correct next form. Its top-1 changes on 77.93% of paired counterfactuals, yet changes to the correct target only 12.89% of the time. Context sensitivity alone is not understanding. The language stage is rejected, writer training is not authorized by the fixed path, and no frozen evaluation is opened.

For diagnosis only, a separate command explicitly marks itself exploratory and continues from the rejected language checkpoint. It freezes the causal field and trains the 7,073,617-parameter continuous flow writer for 1,200 updates. For clean target x , noise ϵ , and time τ it learns the velocity of $y_\tau = (1 - \tau)x + \tau\epsilon$, conditions on h_t and $\hat{z}_{t+1}$, and rereads its sampled pixels. This run cannot change the evidence verdict. On 256 targets and 16 autonomous continuations, generated identity top-1 is zero, reread target cosine is 0.08015, and pixel F1 is 0.32210. Blank rate is zero and position-16 density is 0.97657 times target density. The writer therefore learns ink occupancy, not the intended continuation.

V25 development measure	Measured	Required	Result
Full-history top-1	0.01123	diagnostic	–
Last-only top-1	0.00146	diagnostic	–
Shuffled-history top-1	0.00342	diagnostic	–
Image-unigram top-1	0.01611	diagnostic	below
Symbolic-bigram top-1	0.12158	benchmark	below
Full minus last top-1	+0.00977	> 0.030	fail
Full minus unigram top-1	−0.00488	> 0.030	fail
Full minus last log probability	+0.04882	> 0.050	fail
Full minus shuffled top-1	+0.00781	> 0.015	fail
Counterfactual switch accuracy	0.12891	> 0.550	fail
Full-history target cosine	0.27513	> 0.550	fail
Student boundary / peak CUDA	clean / 0.598 GiB	clean / < 18	pass

Table 9: V25 fixed natural-Chinese development audit. Full history has a small advantage over last-only and shuffled history, but remains below the image unigram and symbolic bigram. Six semantic and causal gates fail.

V25 localizes the next problem to the predictive state rather than page geometry. Cross-font visual appearance, exact identity, and context-predictive meaning share one target state; cosine can improve toward a frequent visual centroid without selecting the next glyph. A subsequent preregistered model should separate an exact appearance field from a context-predictive residual and require both under matched state-shuffle interventions.

Long context has a separate exact representation. For a lattice width K , sequence index t maps serpentine-wise to

$$r = \lfloor t/K \rfloor, \quad c = \begin{cases} t \bmod K, & r \text{ even,} \\ K - 1 - (t \bmod K), & r \text{ odd.} \end{cases}$$

This bijection preserves adjacency at row turns and maps clean $N \times 1 \times 32 \times 32$ cells to either an inspectable flat page or a learned $C \times R \times K$ retinal lattice with a validity mask. A 256×256 lattice holds 65,536 cells, but was not used in V25. Scaling this representation before repairing the 64-cell objective would increase context without establishing better language.



5.9 Factorized Visual Context V26: changed state, failed binding

V26 tests the repair proposed by V25 while preserving the same ordinary-Chinese corpus, record-group partitions, disjoint font partitions, 64-cell horizon, frozen V16 retina, and strict student boundary. It changes the supervised unit. Every natural training example predicts only its full-context endpoint and future offsets 2, 4, and 8; losses are no longer averaged over all short prefixes. A fixed set of 16,384 cross-record training pairs shares the same four-character suffix but has different earlier histories and different next glyphs. Pair suffixes are rendered with identical font and augmentation draws, so their final four image tensors are exactly equal.

For $X = (x_1, \dots, x_{64})$, the appearance route receives only the last visible image and the history route receives only the preceding images:

$$a = A_\theta(R(x_{64})), \quad r = C_\theta(R(x_{1:63})), \quad s = \text{RMSNorm}(a + \sigma(G_\theta([a; r])) \odot r).$$

Zeroing r is therefore a true last-only intervention. For two suffix-matched contexts, swapping r changes only earlier-history state while preserving the last-glyph appearance route.

For horizon $h \in \{1, 2, 4, 8\}$ and independent Gaussian noise ϵ_k , $k = 1, \dots, 8$, a shared conditional head emits unit retinal particles

$$q_{h,k} = \text{normalize } P_\theta(s + e_h + N_\theta(\epsilon_k)) \in \mathbb{S}^{191}.$$

These samples are continuous predictions, not vocabulary rows or persistent glyph prototypes. For target state y_h , the empirical energy score is

$$\mathcal{E}(Q_h, y_h) = \frac{1}{8} \sum_k \|q_{h,k} - y_h\|_2 - \frac{1}{128} \sum_{k,j} \|q_{h,k} - q_{h,j}\|_2.$$

Training combines next- and multi-horizon energy scores, continuous candidate ranking against an 8,192-entry FIFO of detached image states, and a softplus pair-ranking loss with fixed margin 0.10. The queue contains image observations but no IDs and is absent from deployed inference. This design follows the calibrated continuous-prediction motivation of proper energy scores [32]; it does not assume that particle diversity itself is semantic.

The complete model contains 19,142,721 parameters, of which 16,476,865 are trainable. Its fixed run performs 8,000 BF16 AdamW updates in 1,863.27 seconds on one RTX 4090, with 0.888403 GiB peak allocated CUDA memory. The corpus manifest SHA-256 is 76048753...f6866c03, the preregistered protocol SHA-256 is dcfa5b97...b899df1, and the final checkpoint SHA-256 is 065f84e1...db6e1d. The frozen partition is counted but never rendered.

The fixed natural audit uses 2,048 development windows and the same evaluator-only 1,024-form image bank for every branch. Table 10 shows the preregistered mechanism and language comparisons.

The retina-bank oracle is exactly 1.0, so the frozen visual sensor and evaluator can distinguish every audited target. The model nevertheless assigns almost no top-1 advantage to history. Its full-history target log probability is -6.932756 , versus -6.952485 for suffix-4 and -6.936448 after shuffling cells 1–60 while preserving the suffix. Only the full-versus-last log-probability gate passes.

The primary causal audit uses 512 cross-record suffix-4 pairs. Its result is shown in Table 11.

The pair result localizes the failure. The history branch is not constant: pixel-identical suffix contexts produce substantially different residuals. Those differences do not bind to the correct next-glyph distribution. Because suffix-matched pair members have exactly the same appearance state, ordinary two-way pair ranking and the residual-swap target test are algebraically equivalent in this implementation; both are retained to preserve the preregistered intervention interface rather

V26 natural development measure	Measured	Required	Result
Full-history top-1	0.000488	diagnostic	–
Last-only / suffix-4 top-1	0.000488 / 0.000488	diagnostic	–
Shuffled-prefix top-1	0.000488	diagnostic	–
Image-unigram top-1	0.014160	control	below
Symbolic-bigram top-1	0.135254	benchmark	below
Full minus last top-1	0.000000	> 0.020	fail
Full minus unigram top-1	−0.013672	> 0.030	fail
Full minus last log probability	+0.170451	> 0.10	pass
Full minus suffix-4 log probability	+0.019729	> 0.03	fail
Full minus shuffled log probability	+0.003692	> 0.03	fail
Full target cosine / particle spread	0.160451 / 1.210599	diagnostic	–
Retina oracle / student boundary	1.0 / clean	≥ 0.99 / clean	pass
Peak allocated CUDA	0.888403 GiB	< 18 GiB	pass

Table 10: V26 fixed natural-Chinese development audit. Full history improves target log probability over last-only, but not over the same four-cell suffix or suffix-preserving history shuffle. It remains below image unigram and symbolic bigram controls.

V26 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Mean appearance-state difference	0.000000	exact	pass
Mean history-residual difference	4.628258	diagnostic	changed
Correct two-way pair ranking	0.500000	> 0.65	fail
Swapped-residual target accuracy	0.500000	> 0.65	fail
Mean correct-versus-other margin	0.0000677	diagnostic	near zero
Top-1 output switch rate	0.212891	diagnostic	–
Both pair targets top-1 correct	0.025391	diagnostic	–

Table 11: V26 pixel-identical suffix intervention. Earlier history materially changes the continuous residual, but neither ordinary nor swapped-residual scoring prefers the associated target above chance.

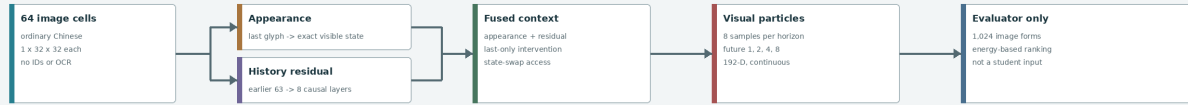
than counted as independent evidence. A suffix-8 diagnostic agrees, with 0.501953 ranking and swap accuracy.

V26 therefore rejects this factorization and objective, not image-native language modeling in general. A broad particle cloud and a changing hidden state are not evidence of useful conditional language. The next attributable experiment should remain at 64 cells and make matched target choice conditionally identifiable during optimization. It must beat suffix-only, history-shuffled, image-unigram, and symbolic-bigram controls before a pixel writer, serpentine page lattice, geometric depth, or motion is authorized.

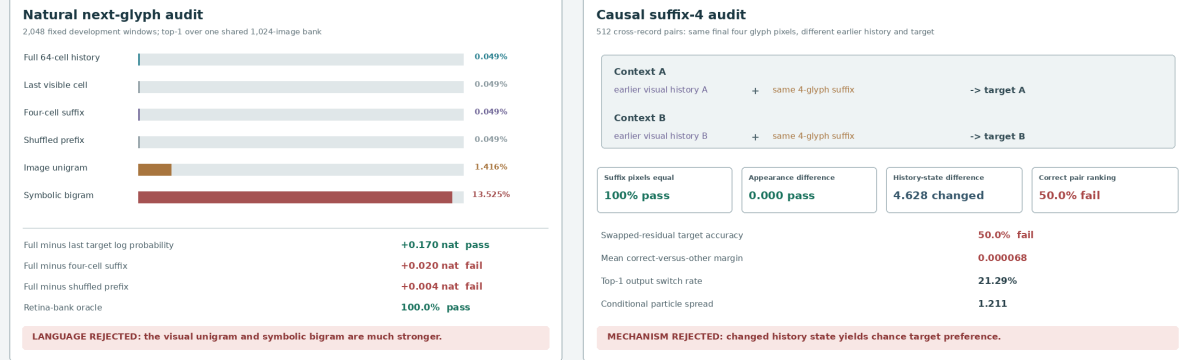
We further localize this failure with a post-hoc diagnostic that cannot revise the V26 result or authorize frozen evaluation. Every V26 parameter is frozen. Three visual scorers, totaling 1.11M trainable parameters, receive appearance, history residual, or fused state and score two arbitrary candidate images using normalized projected compatibility. One pass over all 16,384 train suffix pairs optimizes symmetric 2×2 assignment. The fixed 512 development pairs yield 1,024 cross-font assignments and 2,048 scored arms. Appearance-only accuracy is exactly 0.500000, history accuracy is 0.506836, and fused-state accuracy is 0.503418. Their mean margins are 0.000108 and 0.000371. In contrast, direct frozen-retina cosine assigns the same cross-font target images at 0.999512 with

V26: visual history changes state, but does not bind the next glyph

19.14M total parameters | 8,000 fixed BF16 updates | one RTX 4090 | 0.888 GiB peak | frozen split sealed



The appearance and history routes are separable. Suffix pairs keep the final four glyph images exactly equal while changing earlier text.



What the fixed experiment establishes

The full factorized image-only system trains in 31.1 minutes with 0.888 GiB peak allocated VRAM.
 The retina/evaluator is intact, the exact suffix intervention holds, and earlier history changes the residual state.

What V26 falsifies

Appearance/residual factorization plus energy, queue contrast, and pair ranking does not learn useful next-glyph binding.
 Internal-state sensitivity and broad visual particles are not conditional language behavior.

Next controlled question

Make target choice conditionally identifiable before adding a writer, longer context, page geometry, depth, or motion.
 A next model must beat suffix, shuffled-history, unigram, and bigram controls under paired causal intervention.

DEVELOPMENT FALSIFICATION FROZEN SEALED

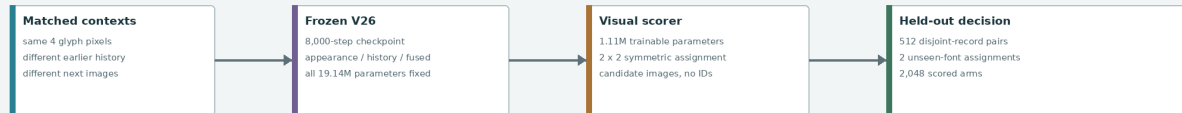
Figure 14: Measured V26 development result generated directly from the fixed audit. The visual retina and exact suffix intervention pass, and earlier history changes the residual state, but natural next-glyph accuracy remains below frequency controls and matched target preference is exactly chance. The frozen partition stays sealed and no writer is trained.

a mean margin of 0.747237. Thus candidate visibility is intact, but V26's history state does not expose a transferable next-glyph relation to this deterministic nonlinear decoder.

The resulting V27 direction is therefore stricter than replacing an output head. A trainable image retina and causal context field must be optimized jointly with deterministic candidate compatibility under matched alternatives. The primary score remains image-to-image and vocabulary-free; an appearance-only chance arm, raw-retina identity control, prefix shuffle, unigram, and symbolic bigram remain required controls. Stochastic form generation follows only after this relation passes.

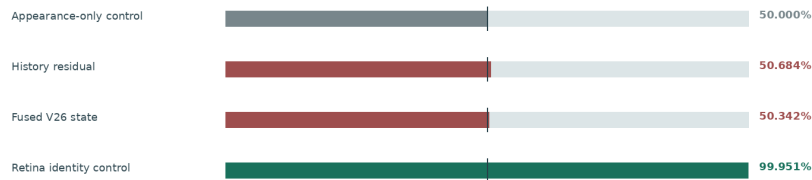
Frozen V26 state does not expose next-glyph binding

Post-hoc localization diagnostic | development only | frozen split sealed | not model evidence



Cross-font paired accuracy

Vertical marker: chance. The positive control uses the same candidate images and frozen retina.



Diagnosis

History changes numerically, but a deterministic nonlinear scorer cannot decode a transferable next-glyph relation from it. Candidate visibility is not the bottleneck.

History margin 0.000108 | Fused margin 0.000371 | Retina margin 0.747237

V27 decision: jointly train the image context encoder and candidate-conditioned compatibility objective before attaching a stochastic visual writer.

One train-pair pass | 93.46 s | 2.482 GiB peak on one RTX 4090

Figure 15: Post-hoc frozen-V26 localization. The raw retina nearly perfectly distinguishes paired target images across unseen fonts, while deterministic candidate scorers over frozen history and fused states remain at chance. This is a development diagnostic, not preregistered model evidence.

5.10 Joint Visual Compatibility V27: visible candidates, failed relation

V27 executes that joint test without changing the corpus, record partitions, font separation, 64-cell horizon, or student boundary. Its online retina initializes from V16 and is optimized at one tenth the main learning rate. An eight-layer width-384 causal rotary field maps the ordered context images to a normalized query. An exponential-moving-average retina and residual image projector map an arbitrary candidate image to a normalized key:

$$q(X) = \text{normalize } Q(F(R_o(X))_{64}), \quad k(Y) = \text{normalize } G_t(R_t(Y)),$$

$$s(X, Y) = \exp(\alpha) q(X)^\top k(Y), \quad \exp(\alpha) \leq 100.$$

The candidate collection is not part of the model or checkpoint. At deployed inference the scorer accepts any floating candidate image. Every paired training view independently permutes its two candidate images, and the evaluator undoes only that visible-array permutation. This makes fixed diagonal position unusable as an identity channel.

The fixed objective combines multi-positive cross-font natural-image contrast, symmetric row-and-column assignment on suffix-matched pairs, cross-font candidate identity, and VICReg variance/covariance regularization:

$$\mathcal{L} = \mathcal{L}_{\text{natural}} + 2\mathcal{L}_{\text{pair}} + 0.5\mathcal{L}_{\text{identity}} + 0.05\mathcal{L}_{\text{VC}}.$$

No vocabulary classification, symbolic target, reconstruction decoder, stochastic particle head, or writer is trained. The complete model has 18,599,553 parameters, of which 17,118,401 are trainable. The single fixed run performs 8,000 BF16 AdamW updates. Training and the complete development audit take 2,347.47 seconds on one RTX 4090 with 2.267916 GiB peak allocated CUDA memory. The protocol SHA-256 is 0c386dd2...cb223310; the final checkpoint SHA-256 is 49dc07f2...1e77e539. The frozen partition is never rendered.

The natural audit uses 2,048 development windows and the same evaluator-only 1,024-image bank for every branch. Table 12 reports the fixed language comparisons.

V27 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.016113 / 0.023926	diagnostic	–
Last-only / suffix-4 top-1	0.001953 / 0.008301	diagnostic	–
Shuffled-prefix top-1	0.013184	diagnostic	–
Image-unigram top-1	0.020508	control	below
Symbolic-bigram top-1	0.125977	benchmark	below
Full minus unigram top-1	−0.004395	> 0.03	fail
Full minus bigram top-1	−0.109863	> 0.01	fail
Full minus bigram log probability	−1.856770	> 0.05	fail
Full minus suffix-4 log probability	+0.059248	> 0.03	pass
Full minus shuffled log probability	+0.002726	> 0.03	fail
Learned 1,024-bank cross-font identity	0.948730	≥ 0.99	fail
Student boundary / peak CUDA	clean / 2.267916 GiB	clean / < 18	pass

Table 12: V27 fixed natural-Chinese development audit. Full context improves target log probability over the four-cell suffix, but the advantage almost vanishes when the same prefix images are shuffled. The model remains below image unigram and symbolic bigram controls.

Only one of five natural-language gates passes. Full target log probability is −6.876778, versus −6.936025 for suffix-4 and −6.879503 after shuffling cells 1–60. Thus the apparent context gain is

not attributable to the order of the earlier writing. Full top-1 is also lower than both frequency controls.

The primary causal audit contains 512 cross-record suffix-4 pairs in two unseen development fonts, yielding 1,024 assignments and 2,048 scored arms. Table 13 reports the fixed mechanism controls.

V27 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Candidate permutation score error	0.000000	$< 10^{-5}$	pass
Candidate permutation agreement	1.000000	exact	pass
Raw V16 two-candidate identity	0.999512	≥ 0.99	pass
Learned 1,024-bank identity	0.948730	≥ 0.99	fail
Full-context arm accuracy	0.507080	> 0.65	fail
Full both-correct rate	0.088867	> 0.40	fail
Last-only / suffix-4 accuracy	0.500000 / 0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.505615	control	near chance
Full minus suffix-4 accuracy	+0.007080	> 0.15	fail
Full minus shuffled accuracy	+0.001465	> 0.05	fail
Full minus shuffled margin	+0.002265	> 0.02	fail

Table 13: V27 pixel-identical suffix intervention. Candidate visibility, permutation equivariance, suffix equality, and chance controls pass. The learned context/candidate relation remains at chance and does not materially exceed a suffix-preserving prefix shuffle.

The raw V16 retina reaches 0.999512 cross-font target identity with 0.736124 mean cosine margin. Candidate pixels and the evaluator are therefore adequate. This is a two-candidate pair control. The learned identity gate instead uses 1,024 candidates, reaches 0.948730, and therefore cannot be subtracted from the raw number as a before/after change. Full assignment remains only 0.1465 percentage point above the shuffled-prefix arm. Seven of thirteen mechanism gates pass, but all five context-binding gates and the learned full-bank identity gate fail.

V27 rejects this global query/key compatibility mechanism, not image-native language modeling in general. It also removes a tempting explanation for V26: the main failure is not that arbitrary candidate images are invisible to the retina. The next controlled experiment should preserve the raw retinal geometry exactly and supervise dense, explicitly ordered future visual fields rather than compressing all language evidence into one global dot product. The authoritative representation remains an ordered $N \times 1 \times 32 \times 32$ stream of full glyph images. A reversible 2D fold may expose parallel spatial computation, but must invert exactly and pass an order intervention. A writer, longer lattice, geometric depth, and motion remain unauthorized until ordered context beats suffix, shuffle, unigram, and bigram controls.

V28 executes this dense-future test below without changing the corpus, partitions, student boundary, or frozen-data rule.

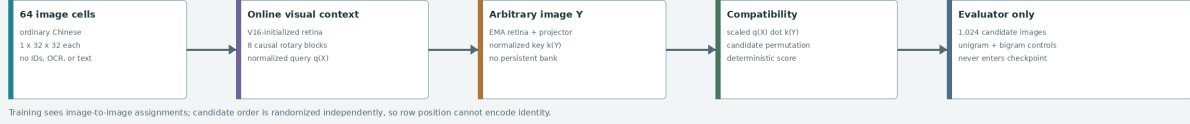
5.11 Dense Visual Future Energy V28: order signal without binding

V28 consumes an authoritative ordered volume $X \in \mathbb{R}^{64 \times 1 \times 32 \times 32}$ and predicts future writing images at offsets $d \in \{1, 2, 4\}$. The fixed V16 retina is never updated. For image x , a residual semantic adapter, initialized as the identity, and its stop-gradient EMA copy produce

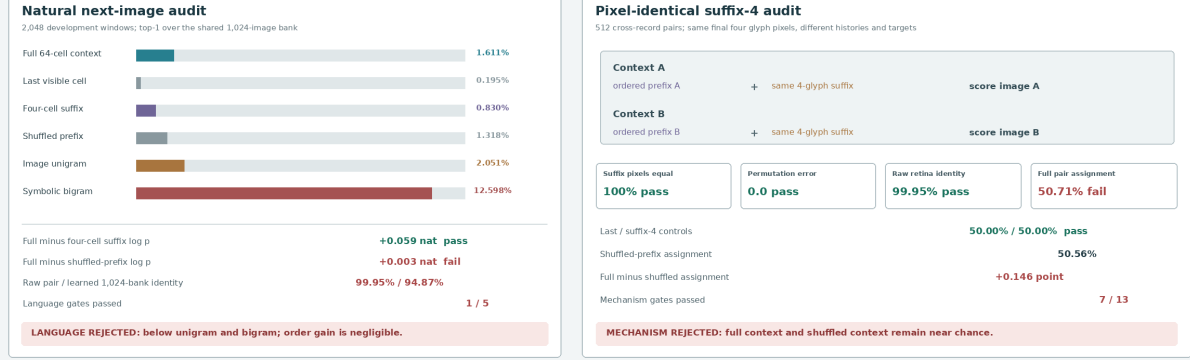
$$r(x) = \text{normalize } R_0(x), \quad z(x) = \text{normalize } A_\theta(r(x)), \quad \bar{z}(x) = \text{EMA}(z(x)).$$

V27: candidate images remain visible, but visual compatibility is not language

18.60M total parameters | 8,000 fixed BF16 updates | one RTX 4090 | 2.268 GiB peak | frozen split sealed



Training sees image-to-image assignments; candidate order is randomized independently, so row position cannot encode identity.



What the fixed experiment establishes

The image-only scorer is permutation-equivariant, the exact suffix intervention holds, and raw candidate visibility is 99.95%. The complete audit runs in 39.1 minutes with 2.268 GiB peak allocated VRAM; frozen writing stays sealed.

What V27 falsifies

Jointly adapting one global context query and one candidate-image key does not learn useful ordered next-writing compatibility. The separate learned 1,024-bank identity gate is 94.87%; full context is only 0.15 point above a prefix shuffle.

Next controlled question

Preserve the raw retina exactly and train dense, explicitly ordered future-field prediction over the full glyph-image stream. Require a causal advantage over suffix and shuffled history before adding a writer, longer lattice, depth, or motion.

DEVELOPMENT FALSIFICATION FROZEN SEALED

Figure 16: Measured V27 development result generated directly from the fixed audit. The image-only boundary, candidate permutation, exact suffix intervention, and raw-retina identity controls pass. Jointly learned visual compatibility remains near chance, misses its separate full-bank identity gate, and does not use prefix order. The frozen partition stays sealed and no writer is trained.

An eight-layer width-384 causal rotary field reads the concatenated raw and semantic states. At each selected position and horizon it emits four normalized raw hypotheses, four normalized semantic hypotheses, and mixture logits:

$$H_{1:t} = F([r(x_1); z(x_1)], \dots, [r(x_t); z(x_t)]),$$

$$\{q_{d,k}^r, q_{d,k}^z, \pi_{d,k}\}_{k=1}^4 = P_d(H_t).$$

The score of an arbitrary candidate image Y is continuous and contains no vocabulary table:

$$s_d(X, Y) = \log \sum_{k=1}^4 \pi_{d,k} \exp\left(\tau_r q_{d,k}^r r(Y) + \tau_z q_{d,k}^z \bar{z}(Y)\right).$$

Each natural microbatch samples 16 stratified causal positions, including the endpoint, and uses opposite-font targets at horizons 1, 2, and 4. The fixed objective is

$$\mathcal{L} = \mathcal{L}_{\text{dense}} + 0.25\mathcal{L}_{\text{ES}} + 0.5\mathcal{L}_{\text{id}} + 2\mathcal{L}_{\text{pair}} + \mathcal{L}_{\text{natural-order}} + \mathcal{L}_{\text{pair-order}}.$$

Here $\mathcal{L}_{\text{dense}}$ is weighted multi-positive cross-font contrast, $\mathcal{L}_{\text{ES}}$ is a strictly proper empirical energy score over the continuous raw hypotheses, and $\mathcal{L}_{\text{id}}$ trains the semantic route across fonts. The final

three terms use identical candidate pixels under symmetric suffix-pair assignment and under explicit full, suffix-only, and suffix-preserving shuffled-prefix interventions. Temporary equality groups are derived from canonical pixels by the host loss; they are not model inputs or inference state.

The complete model has 17,859,142 parameters, of which 16,377,990 are trainable. Its single fixed run performs 10,000 BF16 AdamW updates and the full development audit in 7,134.82 seconds (118.91 minutes) on one RTX 4090, with 1.144400 GiB peak allocated CUDA memory. The protocol SHA-256 is `b8e515d2...a7780f7f`, the final checkpoint SHA-256 is `22503464...2c7f64f`, and the audit SHA-256 is `2cb73707...8f0fb3d`. The frozen partition is never rendered.

The natural audit uses 2,048 development windows and one fixed evaluator-only 1,024-image bank. Table 14 reports the result.

V28 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.014160 / 0.035156	diagnostic	–
Last-only / suffix-4 top-1	0.004395 / 0.015625	diagnostic	–
Shuffled-prefix top-1	0.002441	diagnostic	–
Image-unigram top-1	0.018555	control	below
Symbolic-bigram / trigram top-1	0.131348 / 0.193359	controls	below
Full minus unigram top-1	−0.004395	> 0.03	fail
Full minus bigram top-1	−0.117188	> 0.01	fail
Full minus bigram log probability	−1.801083	> 0.05	fail
Full minus suffix-4 log probability	+0.030369	> 0.03	pass
Full minus shuffled log probability	+0.215112	> 0.03	pass
Raw / EMA 1,024-way identity	0.920410 / 0.964355	EMA $\geq$ 0.95	pass
EMA minus raw identity	+0.043945	$\geq$ 0.02	pass
Student boundary / peak CUDA	clean / 1.144400 GiB	clean / < 18	pass

Table 14: V28 fixed natural-Chinese development audit. Full context changes target probability relative to suffix and shuffled controls, and the semantic route improves same-scope identity. Correct future-image rank remains below the unigram, bigram, and trigram controls.

Full target log probability is -6.829707 , versus -6.860076 for suffix-4 and -7.044819 after shuffling cells 1–60. This is an attributable order effect in probability, but suffix-4 top-1 is slightly higher than full-context top-1 and the symbolic bigram is more than nine times as accurate. Four of six language gates fail.

The primary causal audit uses 512 cross-record context pairs whose final four glyph images are bitwise equal but whose earlier histories and targets differ. Candidate order is independently permuted in two unseen development fonts. Table 15 reports the fixed mechanism controls.

V28 is a clean negative result. It establishes that the $N \times 1 \times 32 \times 32$ visual-time interface, frozen-retina semantic adaptation, dense multi-horizon prediction, and continuous candidate scoring are executable within 1.15 GiB peak allocated memory. It does not establish a useful language model or a capability-normalized efficiency advantage. The failure is now narrower: ordered prefix information changes scores, but the training objective does not distribute that evidence reliably enough to bind each context to its correct future image.

V29 executes that candidate-conditioned incremental-evidence test below without changing the corpus, partitions, student boundary, or frozen-data rule.

5.12 Conditional Visual Density Ratio V29: order without assignment

V29 retains the authoritative image stream $X \in [0, 1]^{B \times N \times 1 \times 32 \times 32}$, where N is reading time. It loads the V28 context input, eight causal blocks, context normalization, frozen V16 retina, and both

V28 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Permutation error / agreement	0.000000 / 1.000000	exact	pass
Raw V16 two-candidate identity	0.999512	diagnostic	–
Full-context arm accuracy	0.495605	> 0.65	fail
Full both-correct rate	0.096680	> 0.40	fail
Last-only / suffix-4 accuracy	0.500000 / 0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.499512	control	chance
Full minus suffix-4 accuracy	−0.004395	> 0.15	fail
Full minus shuffled accuracy	−0.003906	> 0.05	fail
Full minus shuffled mean margin	+0.022070	> 0.02	pass

Table 15: V28 pixel-identical suffix intervention. The full score has a larger mean margin than the shuffled score on a subset of examples, but this movement does not yield reliable correct rank or context-to-target binding.

frozen semantic adapters. V28 future heads are discarded. The causal field returns one retained state per visible image:

$$H_T = C_\theta([r(x_1); z(x_1)], \dots, [r(x_T); z(x_T)]) \in \mathbb{R}^{T \times 384}.$$

An arbitrary candidate image y forms a continuous query $q_0 = W_y[r(y); \bar{z}(y)]$. Two six-head cross-attention layers update this query from all context states. The scalar relation head receives the updated query, its displacement, and its elementwise interaction with the original query:

$$q_{l+1} = \text{EvidenceLayer}_l(q_l, H_T), \quad \rho_\theta(H_T, y) = M_\theta([q_2; q_2 - q_0; q_2 \odot q_0]).$$

There is no candidate-ID embedding or bank-sized output layer.

For prefix P , exact four-image suffix S , and candidate y , define

$$F(P, S, y) = \rho_\theta(H(P, S), y), \quad B(S, y) = \rho_\theta(H(S), y),$$

$$G(P, S, y) = F(P, S, y) - B(S, y).$$

Rows of G are centered over candidates. Natural training contrasts all 1,024 host-side candidate images for full, suffix, and incremental scores and adds an ordered-versus-shuffled increment margin. Exact-suffix pairs add symmetric two-by-two full and incremental assignment, per-row positive margin, and per-row ordered-versus-shuffled margin. Candidate images are encoded once by the frozen image path, but their bank is neither serialized nor required at inference. The student receives no bank index, source string, token, Unicode ID, character ID, OCR result, symbolic probability, or external-model state.

The complete model has 20,080,961 parameters, of which 18,451,585 are trainable. Its single fixed run performs 8,000 BF16 AdamW updates and the full development audit in 4,956.14 seconds (82.60 minutes) on one RTX 4090, with 3.356465 GiB peak allocated CUDA memory. The protocol SHA-256 is `4cb17b79...9ffad0f`, the final checkpoint SHA-256 is `a8ec9919...d508b8a`, and the audit SHA-256 is `16645844...4a5108`. Frozen images are never instantiated.

The fixed natural audit uses 2,048 development windows and one evaluator-only 1,024-image bank. Table 16 reports the result.

The primary causal audit uses 512 cross-record pairs whose final four writing images are bitwise equal while earlier histories and next images differ. Candidate order is independently permuted in two unseen development fonts. Table 17 reports the fixed mechanism controls.

V28: dense visual futures learn form and order, but fail language choice

17.86M total / 16.38M trainable | 10,000 BF16 updates | one RTX 4090 | 1.144 GiB peak | frozen split sealed

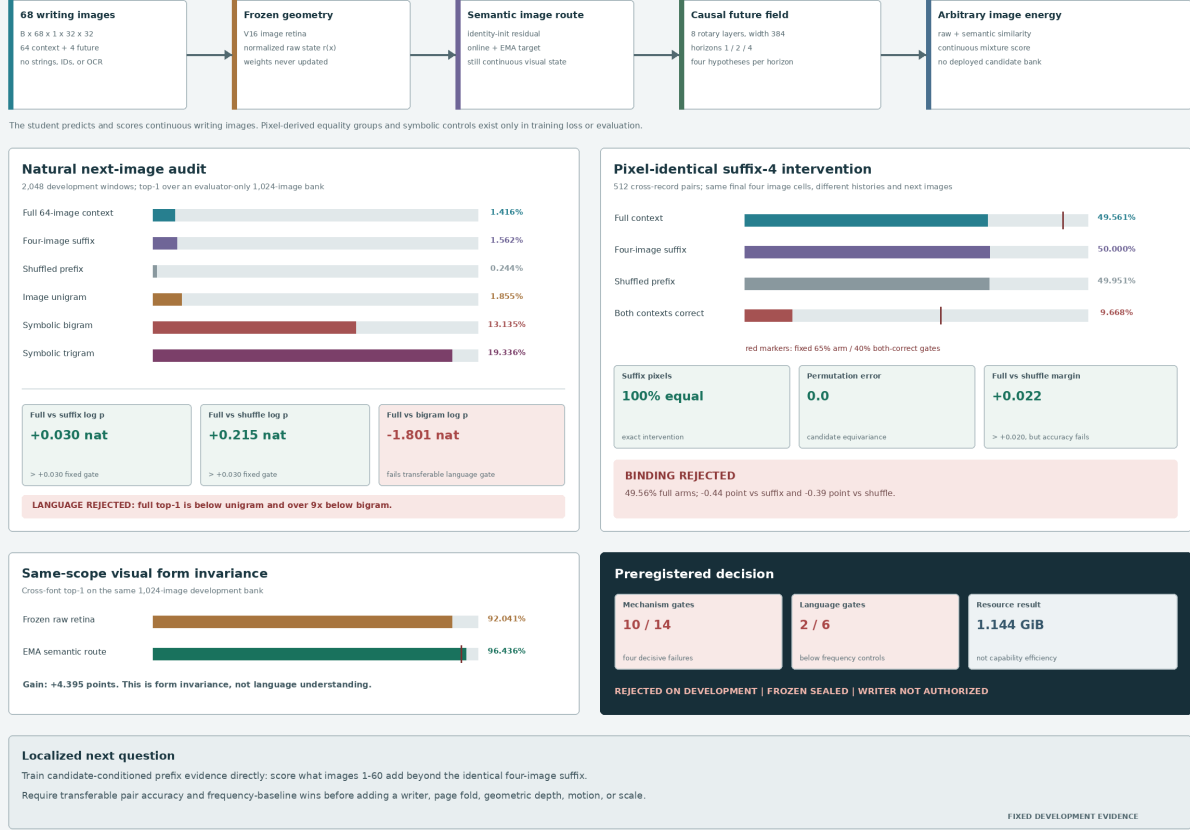


Figure 17: Measured V28 development result generated directly from the fixed audit. Frozen raw geometry and the learned semantic route preserve candidate visibility, and ordered history affects probability and mean margin. Natural selection remains below frequency controls and matched pair assignment stays at chance. Ten of fourteen mechanism gates and two of six language gates pass, so frozen evaluation and writer training remain unauthorized.

The exact suffix exposes a structural limit of the subtraction. Both context rows share one baseline vector (b_1, b_2) . For the diagonal assignment, the mean incremental margin is

$$\begin{aligned} & \frac{1}{2}[(G_{11} - G_{12}) + (G_{22} - G_{21})] \\ &= \frac{1}{2}[(F_{11} - b_1 - F_{12} + b_2) + (F_{22} - b_2 - F_{21} + b_1)] \\ &= \frac{1}{2}[(F_{11} - F_{12}) + (F_{22} - F_{21})]. \end{aligned}$$

The suffix terms cancel exactly, and row centering leaves within-row margins unchanged. Subtraction can redistribute margin between rows, but cannot improve the aggregate assignment margin beyond the full critic. The fixed audit confirms that full and incremental mean margins are both 0.005103277.

V29 is a clean negative result. It rejects this candidate-query critic and density-ratio loss, not image-native language modeling or the ordered visual cell interface. The failure is now more specific: a global scalar critic can learn whether a history resembles natural order without learning which candidate image belongs to which history. A bounded successor should predict a continuous next-

V29 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.023438 / 0.063965	top-1 ≥ 0.15	fail
Suffix-4 / shuffled top-1	0.013672 / 0.011719	diagnostic	–
Increment / shuffled increment top-1	0.023438 / 0.008789	diagnostic	–
Image-unigram top-1	0.013672	control	full above
Symbolic-bigram / trigram top-1	0.138672 / 0.202637	controls	full below
Full minus unigram top-1	+0.009766	> 0.03	fail
Full minus bigram top-1	−0.115234	> 0.01	fail
Full minus bigram log probability	−1.516622	> 0.05	fail
Full minus suffix-4 log probability	+0.179446	> 0.03	pass
Full minus shuffled log probability	+3.102479	> 0.03	pass
Raw / frozen-semantic identity	0.920410 / 0.964355	semantic ≥ 0.95	pass
Boundary / bank absence / peak CUDA	clean / clean / 3.356465 GiB	required	pass

Table 16: V29 fixed natural-Chinese development audit. Earlier ordered images strongly affect target probability, but full 1,024-way ranking remains far below a symbolic bigram. Order detection is not conditional visual choice.

V29 suffix-4 pair measure	Measured	Required	Result
Suffix pixels / score rows	1.000000 / 1.000000	exact	pass
Permutation error / agreement	0.000000 / 1.000000	exact	pass
Raw two-candidate identity	0.999512	≥ 0.99	pass
Full arm / both-correct	0.497314 / 0.038086	> 0.65 / > 0.40	fail
Suffix-only arm accuracy	0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.501221	control	chance
Increment arm / both-correct	0.507080 / 0.089844	> 0.65 / > 0.40	fail
Shuffled-increment arm accuracy	0.495605	control	chance
Increment minus shuffled accuracy	+0.011475	> 0.10	fail
Increment minus shuffled margin	+0.003167	> 0.05	fail

Table 17: V29 pixel-identical suffix intervention. Perception, exactness, and candidate equivariance pass. Full and incremental candidate assignment remain at chance and fail every learned binding threshold.

image patch field from context, encode an arbitrary candidate into the same field, and compare corresponding patches before scalar aggregation. A parameter- and compute-matched bilinear visual scorer should control whether spatial interaction, rather than capacity, provides any gain. The existing suffix, shuffle, frequency, permutation, identity, and boundary controls must pass before a writer, page fold, geometric depth, motion, or larger model is authorized.

V29: visual order signal grows, but candidate binding remains at chance

20.08M total / 18.45M trainable | 8,000 BF16 updates | one RTX 4090 | 3.356 GiB peak | frozen split sealed



Candidate images are scored in chunks. The 1,024-image bank and symbolic controls stay outside model state, checkpoints, and inference.

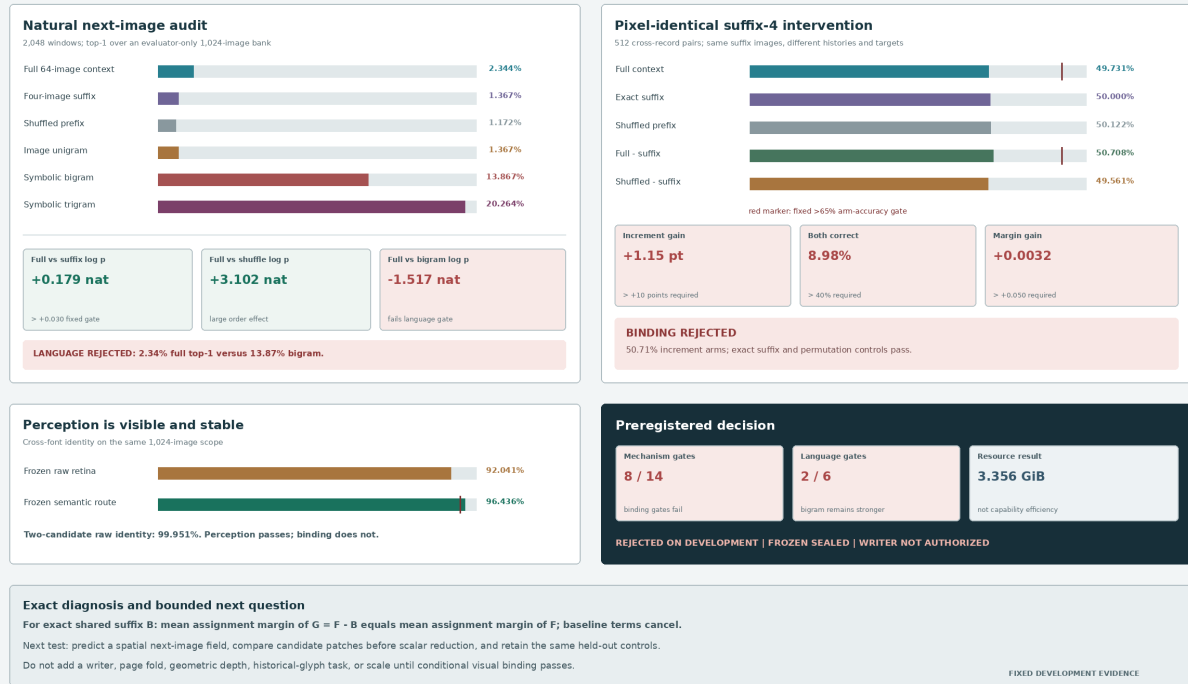


Figure 18: Measured V29 development result generated directly from the fixed audit. The critic detects a large natural order effect while exact-suffix candidate binding stays at chance. All perception, equality, permutation, leakage, and resource controls pass. Eight of fourteen mechanism gates and two of six language gates pass, so frozen evaluation and writer training remain unauthorized.

6 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

For word-origin and book-layout research, 11 additional local works are registered by SHA-256 in `references/source_books/manifest.json`. They total 963,986,234 bytes and include 9,361 PDF pages plus one EPUB: English and Chinese encyclopedias, an English word-origin reference, and Chinese character- origin and oracle-form references. Relative symlinks avoid duplicating or committing the source binaries. Their rights are unverified, so they are private reference, source-comparison, and evaluation material only; they are not part of the releasable training corpus. PDF/OCR extraction may create offline sidecars, but those strings cannot enter a student batch.

7 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t) \mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{V7} = \mathcal{L} + \kappa(t) (0.5 \mathcal{L}_{\text{ctx}} + 0.2 \mathcal{L}_{\text{sample}}).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [36, 35]; it remains image-only because

the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

7.1 PVF state objectives

V8–V16 reuse the proven retina but remove the pixel writer from the state ablation. The causal GRU, proposal, and hyperspherical field are trainable. For proposal μ , target image state z^+ , and image-derived candidate states $\{z_j\}$, visual identity uses

$$\mathcal{L}_{\text{proposal-NCE}} = -\log \frac{\sum_{j \in \mathcal{P}} \exp\{s\mu^\top z_j\}}{\sum_j \exp\{s\mu^\top z_j\}},$$

where positives $\mathcal{P}$ are inferred by frozen retinal similarity. A detached last-image condition supplies the calibrated context objective

$$\mathcal{L}_{\text{proposal-ctx}} = \max\{0, m - \ell(z^+ | h_{\text{full}}) + \ell(z^+ | h_{\text{last}})\}.$$

Equivalent identity and context losses are applied to states numerically sampled from the flow. Direct sampled geodesic loss prevents inference-path drift, while contrastive loss prevents collapse to the retinal mean.

V15 has 10,470,273 total parameters, 9,137,345 trainable parameters, and zero classifier or pixel-actuator parameters. It trains in BF16 with 48-image sequences and peaks at 1.181 GiB allocated CUDA memory. The full V14–V15 trajectory records 1,895.24 seconds on one RTX 4090. Step 2,000 was selected on the development image bank before running the frozen evaluator.

V16 resumes selected V15 step 2,000, adds 6,001,536 causal-memory parameters, and resets the optimizer with a new 100-step warmup. It has 16,471,809 total parameters, 15,138,881 trainable parameters, and zero classifier or actuator parameters. The 1,200-update continuation records 2,047.63 seconds, about 92–114 sequences per second, and 1.479 GiB peak allocated CUDA memory. Step 2,200 was selected before the frozen evaluator by maximum development proposal top-1 subject to full-context over last-only, positive normalized context gain, and target cosine above 0.10.

8 Image-First Training and Inference

Figure 19 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Figure 20 shows the user-facing path. The interface can behave like a normal LLM prompt box. Typed text is rasterized into a clean prompt band; an optional book page, inscription, or glyph crop is normalized and placed beside or below it on the same canvas. Borders, whitespace,

Training ILM-V: Language as Visual Corpus

Every source is converted into page/glyph images. The model learns to predict and generate visual latents, not BPE or word tokens.

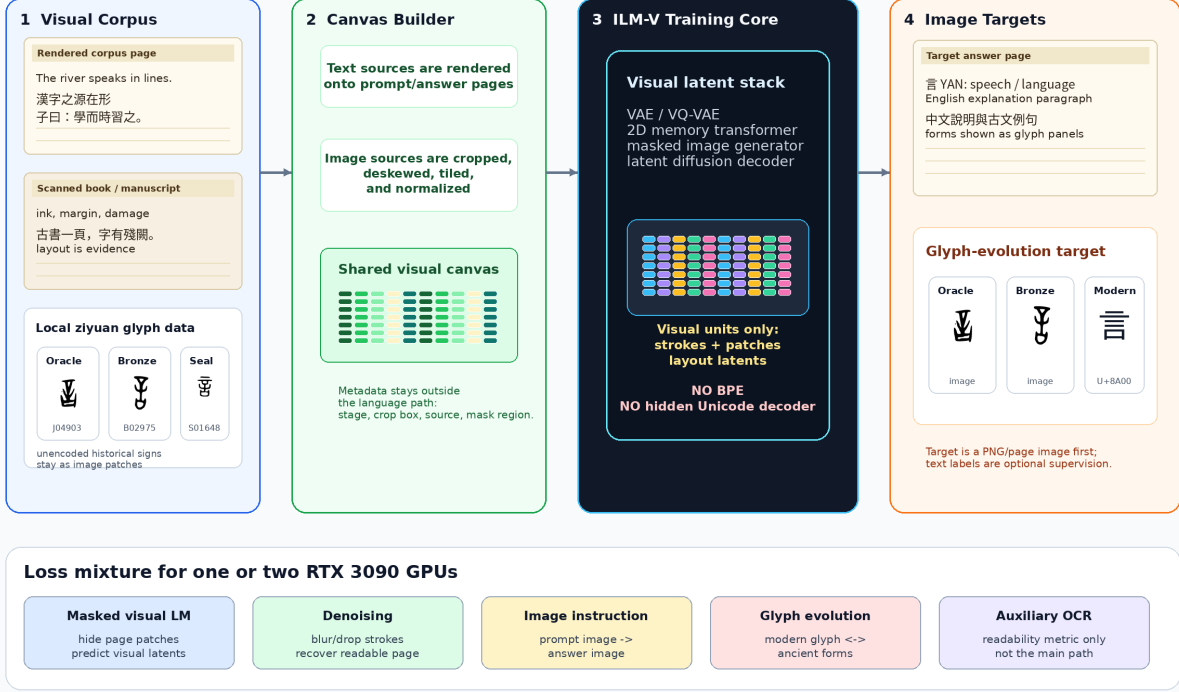


Figure 19: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

and coordinates make the relation visible rather than encoding it in hidden IDs. The ILM can thus answer a typed question alone or a grounded question such as “What does the marked form mean, and how did it evolve?” about an attached page. It returns a page image containing modern language and historical glyph regions. A later OCR/transcoder may attach a searchable layer for encodable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$\begin{aligned}
 r_{\alpha}(\text{text prompt}) &\rightarrow x_{\text{text}}, \\
 g_{\beta}(\text{uploaded page or glyph}) &\rightarrow x_{\text{source}}, \\
 \text{Compose}(x_{\text{text}}, x_{\text{source}}) &\rightarrow x_{\text{prompt}}, \\
 f_{\theta}(x_{\text{prompt}}) &\rightarrow y_{\text{page}}, \\
 o_{\psi}(y_{\text{page}}) &\rightarrow \text{optional text layer} + \text{glyph-region metadata}.
 \end{aligned}$$

Here r_{α} and Compose are deterministic UI adapters, not language components. Either visual region may be absent. The core f_{θ} receives the resulting pixels and maps them to answer image y_{page} . The optional reader o_{ψ} may recover coded text where possible, but unencoded historical forms remain valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

Inference ILM-V: Chat-Like UI, Image-First Output

Typed text and uploaded images are both turned into visual prompt canvases. The model returns a rendered page image first.

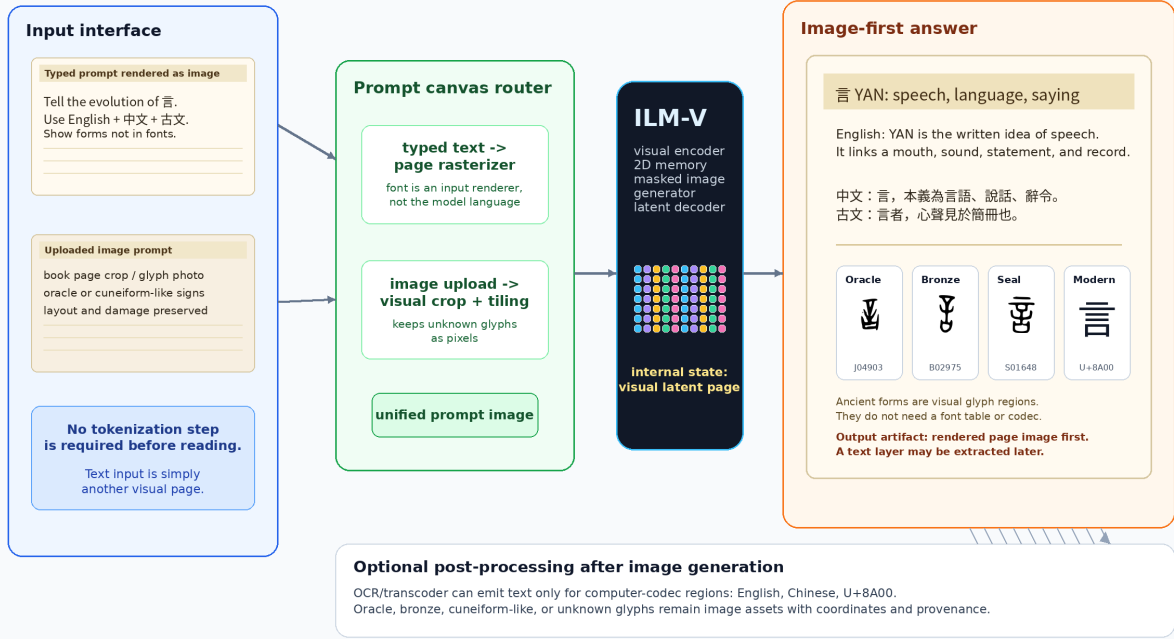


Figure 20: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from “text answer plus illustrations”. The generated page image is the primary output object, and the text layer is secondary.

9 Single-4090 Curriculum

Figure 21 records the broad curriculum. The executable path is intentionally simpler than the complete research space:

1. Read 32×32 writing images into one global state and one small 4×4 spatial field. V16 already provides the retina and causal state.
2. Predict the next continuous visual state from image history and require it to beat last-only, unigram, and bigram controls.

Single-4090 ILM: Minimal Working Path

One visual substrate from ordinary prompts and book pages to generated answer pages

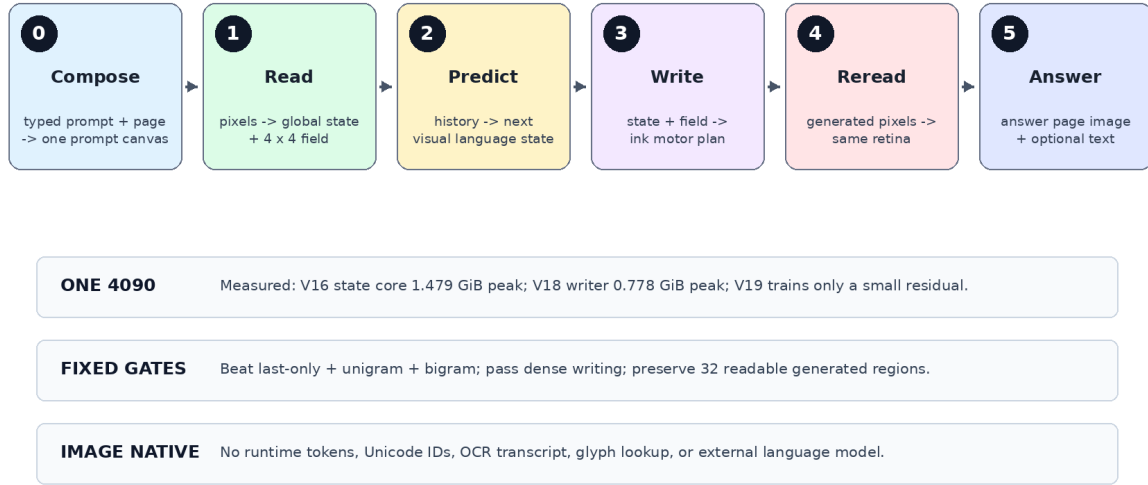


Figure 21: A staged curriculum that keeps each causal proof small enough for one RTX 4090 before any page-scale expansion.

3. Draw that state with a continuity-preserving local route. V19 shows that a small optional residual is ignored; V20 makes field detail causal; V21 makes both occupancy and detail field-causal but rejects disjoint patch rasterization. The successor may overlap only bounded neighboring patches.
4. Render typed prompts and optional source pages into visual frames, then predict local/global answer fields from open question–answer trajectories. Require held-out prompt counterfactuals and prompt-shuffle controls.
5. Feed generated answer pixels back through the same retina and require a readable, prompt-dependent multi-frame autonomous rollout.
6. Add model width or a second GPU only after the preceding fixed gate passes; use the second GPU for throughput or independent replication, not to hide an unresolved objective.

This uses one retina, one causal field, one deterministic motor planner, and one write–reread loop. Stochastic surface generation, billion-parameter scaling, and broad multimodal alignment are deferred until this minimal system works. The same Visual Language Stream can later expose geometric depth and time, but 3D glyph strings and character movies are also deferred until the 2D write–reread gate passes.

The 16.47M PVF V16 state model, 5.73M V17 actuator, and 2.36M V18 motor planner are deliberately small enough for one RTX 4090. Their measured peaks are 1.479, 1.588, and 0.778 GiB, respectively, leaving substantial capacity headroom for controlled corrections. V20 reduces the causal routing test to two exactly matched 0.506M-parameter arms, peaking at 0.323 and 0.397 GiB, and cheaply rejects the writer despite positive field causality. This is not an end-to-end efficiency

Example Output: Evolution of Chinese Character Zhong

A target answer is itself an image: explanation plus historical forms.



Figure uses local hanziyuan-derived glyph files when available; generation targets should retrieve/cite real exemplars.

Figure 22: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

victory: the causal language core remains below a symbolic bigram, dense topology remains imperfect, and autonomous continuation has not passed. “More efficient than an LLM” remains an unsupported hypothesis until quality is matched on a named task.

10 Target Demonstration: Visual Word-Origin Book

The first demo should answer either a normal rendered prompt such as “Explain the evolution of the Chinese character yan” or the same question composed with a photographed book page. It must also support English word-origin questions. The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as “yan is the visible trace of speech”,
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 22 is a compact stage-timeline example; the Yan figures in Figures 1 and 20 show the fuller answer-page style. Together they test whether the model can combine language understanding, visual page composition, and historical script generation.

11 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

AgInTi-Assisted Research Artifact Loop

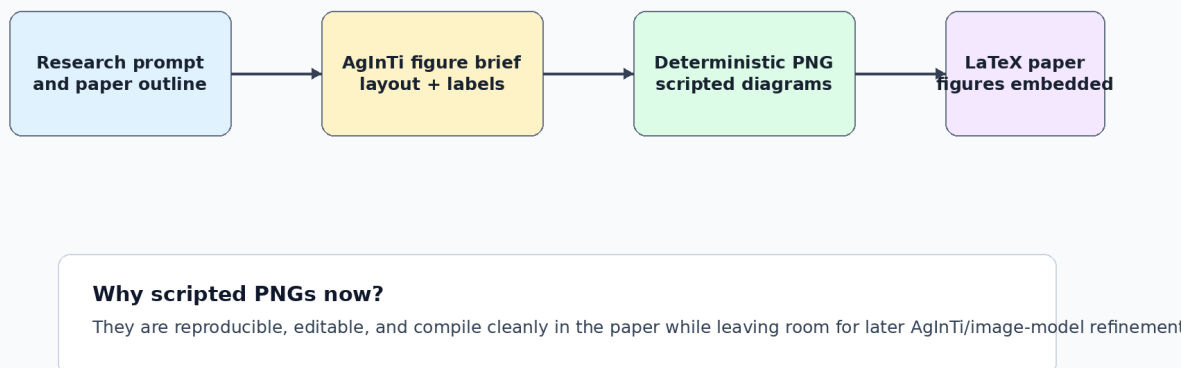


Figure 23: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

12 Evaluation

The model should be evaluated as both a language model and a visual generator:

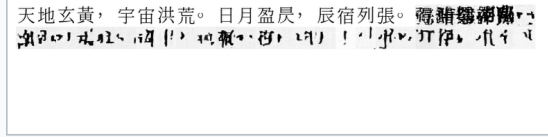
- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.
- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Visual Word-Origin Book tasks: factual proposition accuracy, modern-text readability, prompt-image grounding, attested glyph-stage and source accuracy, unencoded-region preservation, and explicit synthetic-form labeling.
- Optional output transcription scored separately from the native answer image so an OCR sidecar cannot conceal a failed visual answer.

Closed-loop visual trajectory training: what changed

Matched 32-cell autonomous generation; same prompt, seed, sampler, and 11.69M-parameter model

V5 Clean-prefix training

Ink thins under its own visual feedback



Rendered prompt followed by 32 model-generated image cells

Late / early ink

0.483

Sparse cells

37.5%

Frozen-bank top-1

0.908%

Generated context signal

+0.0211

V6 Model-induced rollout training

Ink survives, but the writing remains unreadable



Rendered prompt followed by 32 model-generated image cells

Late / early ink

1.168

Sparse cells

18.8%

Frozen-bank top-1

1.197%

Generated context signal

+0.0077

Measured verdict closed-loop stability improved; contextual language and readable generation did not

Frozen glyph bank SHA-256 543987ddd754135cff34dbec0f6f91ce1971741d0807754c65650031121181c4

Figure 24: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

System	Measured signal	Interpretation
Whole-page flow	velocity MSE 0.851	Unreadable page texture; negative baseline
Global visual memory	top-1 2.23% over 2,016	Above random, but 0/16 historical paraphrases
Causal InkStream	validation ink F1 0.639	Autonomous repeated-grey collapse; negative baseline
RFLM V5 retina oracle	top-1 97.65%	Strong cross-font visual alphabet
RFLM V5 recurrent state	top-1 0.91%	Below last-only 1.36%, unigram 1.86%, and bigram 13.58%
RFLM V6 recurrent state	top-1 1.20%	Improved, but below last-only 1.69% and both language baselines
RFLM V6 pixel writer	context gain +0.0077	Positive but weaker than V5's +0.0211
RFLM V6 autonomous loop	ink ratio 1.168	Stable character-scale marks remain unreadable
RFLM V7 recurrent state	top-1 2.31%	Above last-only 2.02% and unigram 1.86%, far below bigram 13.58%
RFLM V7 calibration	log-probability gain -0.2155	Much better than V6's -0.9066, but full history still lowers mean target probability
RFLM V7 pixel writer	context cosine +0.0303	Passes generated-signal gate; autonomous writing remains unreadable

Same prompt, seed, model size, sampler, and 32-cell horizon

天地玄黃，宇宙洪荒。日月盈昃，辰宿列張。錯羅罔象，攄尋
總覈渺漭，廣廣濶濶，混混沌沌，罔究其極。其列

ink ratio 1.168 sparse 18.75% unreadable

[illegible]

ink ratio 1.050 sparse 15.63% unreadable

Verdict: V7 closes most of the calibrated context deficit, but dense pseudo-glyphs are not readable Chinese. Separate visual-state prediction from rendering.

The fixed evaluation contains 512 common Han candidates with four independent font views and 2,423 eligible held-out contexts. V6 and V7 use the identical bank SHA-256. Selected V7 raises full-context top-1 from 1.197% to 2.311% and top-5 from 3.219% to 5.613%, while the retina oracle remains near 98.27%. Full context now ranks above V7 last-only (2.022%) and unigram (1.857%) but remains far below symbolic bigram (13.578%). Its normalized target-log- probability gain is -0.2155 , improved from a corrected V6 value of -0.9066 but still negative. Raw target-score gain stays positive in both models, showing that raw energy was a misleading metric. The formal acceptance result remains false, and retina-oracle accuracy cannot be cited as language accuracy.

V16 was selected at step 2,200 without reading that frozen bank and evaluated once. Its proposal obtains 6.264% top-1 (112/1,788), versus 3.971% last-only, 1.734% unigram, 0.224% random dynamics, 0.195% chance, and 13.143% symbolic bigram. Its proposal context gain is +0.07732. The stochastic field obtains 3.691% top-1 versus 3.244% last-only, normalized context gain +0.03579, sampled target cosine 0.19506, and sampled context cosine gain +0.07947. V16 adds seven correct proposal contexts over V15, while proposal top-5 and target cosine decline. This is directional evidence and not a statistically established memory advantage; paired predictions on a new frozen bank are required for attribution.

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PVF branch	Frozen signal	Interpretation
V14 proposal	top-1 2.52%	First proposal gate pass; above last-only 2.24% and unigram 1.73%
V14 state flow	top-1 2.35%	First stochastic state gate pass
V15 proposal	top-1 5.87%	Above last-only 4.42%, unigram 1.73%, and random 0.17%; context gain +0.0707
V15 state flow	top-1 3.41%	Above last-only 2.68% and unigram; sampled context cosine gain +0.0805
V16 proposal	top-1 6.26%	112/1,788; above last-only 3.97% and V15 by seven contexts; context gain +0.0773
V16 state flow	top-1 3.69%	Above last-only 3.24% and unigram; sampled context cosine gain +0.0795

Table 18: Frozen Predictive Visual Field results. V15 and V16 branches pass their visual-state gates but remain below the 13.14% symbolic bigram.

V17 frozen actuator metric	Correct state	Shuffled state	Gate
Global identity top-1	58.594%	0.977%	pass
Target cosine	0.71301	0.08612	pass
Pixel F1	0.43849	0.28001	fail (< 0.50)
Ink fraction	0.15947	0.15967	controlled
Human readability	rejected	rejected	fail

Table 19: Untouched V17 evaluation with identical style and noise in each paired branch. Causal visual identity succeeds while readable topology fails.

passes every automatic development topology gate and makes many held-out forms readable, but leaves dense-form and prospective human-readability gates unresolved. V19 then rejects an additive local-field repair: dense F1 is 0.7278, but field shuffling reduces it by only 0.0088 and field removal by only 0.0054. V20 then forces unavailable within-block detail through the field. Correct dense F1 exceeds shuffled-field by 0.1218 and zero-field by 0.3613, with exact quadrant locality, so the local-causality hypothesis passes. Writer selection still fails, and endpoint dense advantage over the equal-capacity control is only 0.0179 against a fixed 0.03 margin. V21 removes the remaining global spatial route. At its best diagnostic snapshot, correct dense F1 exceeds shuffled-field by 0.1739 and zero-field by 0.3465, identity top-1 is 79.10%, target cosine is 0.8331, locality is 1.0, and every exact-basis invariant passes. The tiled-global control collapses to repeated patches. Overall, simple, and medium F1 remain below their fixed gates, so no candidate selects and no paired, human, or frozen stage is authorized. V22 then tests the first prompt-to-answer image stream. Its candidate changes answer identity correctly for only 0.78% of query-counterfactual pairs, versus 0% for the query-blind control, while identity and writer quality are nearly equal. Every candidate selector argmax lands on the operation image. Thus the implemented stream boundary works, but an entropy-sharpened single-frame selector cannot perform the required multi-frame relation. V23 then qualifies and freezes an image canonicalizer before learning the relation. Its candidate reaches 99.80% minimum query/operation switch on a fresh paired audit; the corresponding blind controls are exactly invariant. The opaque visual review and single frozen evaluation also pass, with 99.46% frozen identity top-1 and 0.7848 pixel F1. This turns V22’s interface into the first accepted bounded image-prompt-to-image-answer circuit in the project. V24 then replaces six absolute frame roles with visible headers in a 15–24-frame packet stream and closes a two-frame

V18 development motor-plan metric	Correct intent	Shuffled intent	Gate
Global identity top-1	73.633%	0.977%	pass
Target cosine	0.84618	0.07162	pass
Pixel F1	0.65772	0.31290	pass (> 0.60)
Ink fraction	0.17505	0.17505	controlled
Human readability	simple/medium clear	unrelated	dense forms fail
Frozen images instantiated	no		sealed

Table 20: Fresh V18 development audit with style held fixed and only continuous intent shuffled. Automatic topology gates pass, but frozen promotion is withheld because the human rubric was not numerically prespecified.

visual autoregressive loop. The fresh candidate reaches 99.22% query, 99.32% operation, and 99.61% generated-history switch accuracy; each corresponding blind control is exactly invariant. The opaque audit scores 47/48 and 48/48 for the two output frames. On the single frozen run, frame-1 unseen-identity accuracy is 98.67%, frame-2 label accuracy is 99.73%, and held-out-length scores are 98.47% and 100%. Every fixed gate passes. V25 then removes the packet grammar and exposes the larger gap. Full natural- Chinese visual history reaches 1.123% top-1, above 0.146% last-only and 0.342% shuffled history, but below 1.611% image unigram and 12.158% symbolic bigram. Six fixed gates reject the language model. The exploratory writer’s nonblank images preserve density while obtaining zero identity top-1, so appearance stability cannot substitute for semantic continuation. The frozen partition remains sealed. V26 then separates last-glyph appearance from the preceding visual history and replaces one-point prediction with continuous future particles. It passes the full-versus-last log-probability gate, the perfect retina-oracle gate, the image-only boundary, and the resource gate. It fails the more specific causal tests: full history gains only 0.0197 nat over suffix-4 and 0.00369 nat over a suffix-preserving shuffle. Pixel-identical suffix pairs produce a 4.628 mean history-residual difference but exactly 50% correct target ranking. Thus the representation reacts to history while the predictive distribution does not use that reaction correctly. V26 rejects the proposed factorization and loss; the frozen partition remains sealed and no writer is trained. A post-hoc frozen-state probe further shows that this is not solely a particle- proposal bottleneck: a deterministic nonlinear scorer reaches 50.684% from history and 50.342% from fused state while a raw-retina identity control reaches 99.951%. This motivates jointly learning visual context representation and candidate compatibility rather than preserving V26’s encoder unchanged. V27 performs that joint test. Its full-context pair assignment is 50.708%, versus 50.562% after shuffling the earlier prefix, while natural top-1 is 1.611%, below 2.051% image unigram and 12.598% symbolic bigram. The raw retina still gives 99.951% cross-font identity and candidate permutation error is exactly zero in the two-candidate pair control. Separately, learned 1,024-bank identity is 94.873% and misses its 99% gate; the two retrieval sizes are not directly comparable. V27 therefore rejects the global query/key compatibility route; its frozen partition remains sealed. V28 then freezes the raw retina and distributes supervision across 16 causal positions and three future horizons. The same-scope 1,024-way EMA semantic identity reaches 96.436%, 4.395 percentage points above the raw-retina route, and full target log probability gains 0.215 nat over the shuffled-prefix arm. Nevertheless, natural top-1 is 1.416%, below 1.855% unigram and 13.135% bigram, while full-context pair assignment is 49.561% versus 49.951% after prefix shuffling. The pair mean-margin gate passes, but the rank gates fail. V28 therefore rejects dense global future energy as a sufficient binding mechanism; its frozen partition remains sealed and no writer is trained. V29 then makes each candidate image cross-attend to retained per-cell context

states and trains the centered full-minus-suffix increment directly. Full target log probability gains 3.102 nat over suffix-preserving prefix shuffle, but natural top-1 is only 2.344% versus 13.867% for bigram. Exact-suffix incremental assignment is 50.708%, versus 49.561% shuffled, and both-correct is 8.984%. Raw two-candidate identity is 99.951%, frozen-semantic 1,024-way identity is 96.436%, and all equality and permutation controls pass. V29 thus separates a strong global order effect from candidate-specific binding. It also shows algebraically and empirically that a shared suffix baseline cancels from the aggregate two-by-two assignment margin. The critic itself must learn the missing interaction; changing only the baseline cannot supply it. The combined result establishes that a small image-only student can learn causal language signal, that a small continuous field can control topographic generated ink, and that several visual roles can be composed into a rendered answer stream whose second frame depends on rereading the first. It does not establish useful autonomous language. V25 shows that simply replacing the hand-authored grammar with a generic causal visual field is insufficient. V26 shows that separating appearance from history is also insufficient when the objective permits changed residuals with nearly unchanged target preference. V27 shows that joint dot-product candidate compatibility is also insufficient: it preserves raw target visibility in the pair control but neither passes the separate learned full-bank identity gate nor makes earlier order useful. The V28 shows that freezing that retina, improving semantic identity, and adding dense ordered future supervision can make earlier order affect probability and margin without producing correct binding. V29 shows that candidate-conditioned cross-attention and explicit prefix-increment contrast can amplify the natural order effect while exact-suffix assignment remains at chance. The next proof should make the context construct a continuous spatial next-image field and compare arbitrary candidate patches before scalar reduction, against a matched bilinear scorer. Only after the correct history beats suffix, suffix-preserving shuffle, unigram, and bigram for the same candidate pixels should a serpentine lattice, variable answer length, or multiple generated lines expand the task. This order is also consistent with measured error accumulation in iterative diffusion chains [37].

13 Conclusion

The experiments show that merely replacing tokens with pixels is insufficient. A page flow can learn paper statistics, a causal writer can learn local ink, and a retina can learn character identity without acquiring language dynamics. RFLM contributes a strict test harness for ordered visual reading, direct ink flow, and autonomous rereading, but V6 and V7 reject exposure-bias correction and a monolithic language-plus-rendering writer as sufficient solutions. PVF then factors perception, causal language state, stochastic visual form, and actuation. V16's 6.26% proposal and 3.69% state-flow top-1, positive normalized history gains, and random/unigram/last-only controls replicate the accepted visual-state proof in this project. They show that causal language signal can be learned directly from writing images by a 16.47M model on one consumer GPU. V17 then reaches 58.59% frozen visual-identity retrieval versus 0.98% under shuffled intended states with only 5.73M trainable parameters and 1.588 GiB peak memory. This breaks the categorical claim that lightweight continuous visual intent cannot control generated writing. Its 0.438 pixel F1 and unreadable pseudo-characters reject the stronger claim. V18 then uses only 2.36M trainable parameters and 0.778 GiB peak memory to reach 0.658 pixel F1 and 73.63% identity top-1 on fresh development renderings, versus 0.313 and 0.98% when intent is shuffled. This demonstrates that structured, recognizable writing can be generated as continuous image topology by a compact model on one consumer GPU without token unembedding. Dense forms still fail, the V18 frozen bank remains sealed, and the intended state is supplied rather than predicted. V19 adds a clean

split, fixed complexity strata, five interventions, and a prospective blinded rule, then rejects its own proposed repair: the correct field gains only 0.0088 dense F1 over a shuffled field and 0.0054 over no field. This demonstrates that exposing local features is not equivalent to using them causally. V20 changes the routing constraint with two exactly matched 0.506M-parameter arms. It obtains a 0.1218 dense-F1 field-shuffle gap, a 0.3613 zero-field gap, and 1.0 local-occlusion locality. This establishes necessary, topographic use of the continuous retinal field. V20 is nevertheless rejected as a writer: overall F1, target cosine, exact decomposition, and the paired control margin fail, so no checkpoint, human review, or frozen evaluation is accepted. V21 then makes both coarse occupancy and detail originate from each corresponding local cell, with global state restricted to uniform modulation. Its 0.1739 shuffled-field dense gap, 0.3465 zero-field gap, 79.10% identity, 0.8331 target cosine, 1.0 locality, and exact basis receipts establish the stronger field-complete routing result. The equal-parameter global-only source cannot draw a spatial plan. Yet 0.6038 overall, 0.5648 simple, and 0.5945 medium F1 reject V21’s disjoint-patch writer. No candidate selects, and no paired, human, or frozen claim is accepted. V22 then closes the interface for a bounded six-frame visual prompt and one-frame image answer with two exactly matched 3.410M-parameter arms. It rejects the stronger understanding claim: 0.0078 counterfactual switch accuracy, 0.0089 paired-output L1, and 0.1592 identity top-1 are not meaningfully better than the query-blind control’s 0, 0, and 0.1533. The operation frame receives all 1,024 candidate selector argmaxes. This demonstrates why prompt-dependent image generation requires relation composition, not simply a visual Transformer, sharp attention, and a pixel writer. V23 supplies that composition with a separately qualified writer and three matched 25,602-parameter relation arms. Fresh development controls, an opaque visual review, and the single frozen run agree: the candidate uses both query and operation images, preserves pair-order invariance, generalizes to held-out compositions and 98 unseen identities, and renders the selected form. Frozen binary choice is 0.9983, identity top-1 is 0.9946, and pixel F1 is 0.7848. The maximum supported claim is fixed-grammar visual relation following. The roles and relation algebra remain built into the circuit, and the answer remains a canonicalized visible source form. V24 removes absolute role positions, varies the prompt from 15 to 24 raster frames, and emits two images. Five matched 1,347-parameter arms localize the causal chain: query-, operation-, and history-blind controls each have exactly zero response to their hidden factor, while header blindness destroys role localization. The candidate’s single frozen run over 107 unseen identities reaches 0.9867 frame-1 identity, 0.9973 frame-2 label accuracy, and 0.9961 history switch accuracy; every fixed gate passes. V24 therefore establishes generated-pixel rereading in a bounded variable-input stream. It does not turn the fixed packet algebra into natural language. V25 performs that first ordinary-language test at a bounded 64-cell horizon. Its 1.123% full-history top-1 exceeds last-only by 0.977 percentage points and shuffled history by 0.781 points, establishing a weak ordered dependency. It does not exceed image unigram, misses the log-probability gate by 0.00118 nat, fails counterfactual and target-cosine gates, and remains 10.8 times below the symbolic bigram. The exploratory writer reaches zero generated identity. The 25.55M-parameter system fits easily on one RTX 4090, but capability-normalized efficiency is not established. The frozen partition is not queried. V26 performs the prescribed separation and replaces a point proposal with eight-particle conditional distributions at four future horizons. Its 19.14M-parameter system completes 8,000 BF16 updates in 31.05 minutes with 0.888 GiB peak allocated memory. This is a resource result, not a language result. Full-history top-1 is 0.0488%, below 1.4160% image unigram and 13.5254% symbolic bigram. More decisively, exact suffix matching makes the appearance-state difference zero while earlier histories produce residual difference 4.628; correct pair ranking and swapped-residual target accuracy are still exactly chance. The perfect retina oracle rules out target-identity loss. No mechanism selects, no writer is trained, and the frozen partition is not queried. V27 then jointly trains the context field and deterministic arbitrary-

candidate image compatibility. Its 18.60M-parameter model completes the fixed 8,000 updates and audit in 39.12 minutes with 2.268 GiB peak allocated memory. The raw V16 retina still gives 99.951% cross-font identity, suffix pixels are exact, candidate permutation error is zero, and last-only and suffix-only arms are exactly at chance. Yet full-context assignment is only 50.708%, versus 50.562% after prefix shuffling. Separately, learned 1,024-bank identity is 94.873% and misses its 99% gate; it is not directly comparable to the two-candidate raw control. Natural top-1 is 1.611%, below 2.051% unigram and 12.598% bigram, while its full-versus-shuffled log-probability gain is only 0.00273 nat. V27 passes seven of thirteen mechanism gates and one of five language gates. No frozen query or writer is authorized. V28 performs the dense future-field test with a frozen retina. Its 17.86M-parameter system completes 10,000 BF16 updates and audit in 118.91 minutes with 1.144 GiB peak allocated memory. The EMA semantic route improves same-scope 1,024-way identity from 92.041% to 96.436%, and full context gains 0.0304 nat over suffix-4 and 0.2151 nat over shuffled-prefix target log probability. Yet natural top-1 is only 1.416%, below unigram and bigram, while pixel-identical-suffix pair assignment is 49.561% versus 49.951% shuffled. V28 passes ten of fourteen mechanism gates and two of six language gates. No frozen query or writer is authorized. V29 performs that candidate-conditioned incremental test. Its 20.08M-parameter system completes 8,000 BF16 updates and audit in 82.60 minutes with 3.356 GiB peak allocated memory. Full target log probability gains 0.1794 nat over suffix-4 and 3.1025 nat over shuffled prefix, yet natural top-1 is 2.344% versus 13.867% bigram. Exact-suffix increment assignment is 50.708% versus 49.561% shuffled; both-correct is 8.984%. Full and incremental aggregate mean margins are both 0.005103 because the common suffix baseline cancels. V29 passes eight of fourteen mechanism gates and two of six language gates. No frozen query or writer is authorized. The seven-context V16 increase, V25 ordered-history increase, V26 changed history state, V27 global compatibility score, V28 probability/margin order effect, and V29 density-ratio order effect are therefore all too weak to establish a useful causal memory core. The next attributable test must keep ordinary Chinese and the strict image boundary, preserve the strong visual geometry, and predict a continuous next-image patch field before scoring an arbitrary candidate. A parameter-matched bilinear visual scorer, suffix-only, order-shuffled, visual-unigram, and symbolic-bigram controls must pass before long context or a writer is promoted. The exact serpentine mapping may then fold clean 32×32 cells into a 256×256 retinal lattice without making an 8×8 thumbnail the sole record. Page-native canvases, 3D geometry, color, and motion remain later observable channels and may not encode hidden character IDs. Prompt-paraphrase, fact-substitution, evidence-removal, generated-history-blind, retrieval, and OCR-plus-text controls must distinguish language use from reconstruction and lookup. Only selected components may enter longer visual memory. The named product gate is a Visual Word-Origin Book that reads a rendered prompt alone or composed with a source page, then emits one answer image or an ordered answer-image stream plus an optional declared text sidecar. Prompt counterfactuals must change that answer correctly; input reconstruction and glyph retrieval do not constitute understanding. Historical etymology answers and broad instruction following remain downstream goals, not present capabilities, and no Qwen-8B parity or end-to-end efficiency claim is warranted. The perception–prediction–motor–self-perception factorization may later use the Visual Language Stream for 3D character strings or writing movies, and may be tested on continuous speech or vocal fields. None of those higher-dimensional sensory-language extensions is established here.

References

- [1] Brown et al. Language Models are Few-Shot Learners. <https://arxiv.org/abs/2005.14165>

- [2] OpenAI. GPT-4 Technical Report. <https://arxiv.org/abs/2303.08774>
- [3] OpenAI. GPT-4o API model documentation. <https://developers.openai.com/api/docs/models/gpt-4o>
- [4] OpenAI. Images and Vision API documentation. <https://platform.openai.com/docs/guides/images-vision>
- [5] OpenAI. Image generation guide. <https://platform.openai.com/docs/guides/image-generation>
- [6] Google. Gemini API image generation documentation. <https://ai.google.dev/gemini-api/docs/image-generation>
- [7] Meta AI. The Llama 3 Herd of Models. <https://arxiv.org/abs/2407.21783>
- [8] Kim et al. OCR-free Document Understanding Transformer. <https://arxiv.org/abs/2111.15664>
- [9] Lee et al. Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding. <https://arxiv.org/abs/2210.03347>
- [10] Rust et al. Language Modelling with Pixels. <https://arxiv.org/abs/2207.06991>
- [11] Tai et al. PIXAR: Auto-Regressive Language Modeling in Pixel Space. <https://arxiv.org/abs/2401.03321>
- [12] Chai et al. Autoregressive Pre-Training on Pixels and Texts. <https://arxiv.org/abs/2404.10710>
- [13] Clark et al. CANINE: Pre-training an Efficient Tokenization-Free Encoder. <https://arxiv.org/abs/2103.06874>
- [14] Xue et al. ByT5: Towards a Token-Free Future. <https://arxiv.org/abs/2105.13626>
- [15] Yu et al. MEGABYTE: Predicting Million-byte Sequences with Multiscale Transformers. <https://arxiv.org/abs/2305.07185>
- [16] Pagnoni et al. Byte Latent Transformer: Patches Scale Better Than Tokens. <https://arxiv.org/abs/2412.09871>
- [17] Zheng et al. Scratchpad Patching: Decoupling Compute from Patch Size in Byte-Level Language Models. <https://arxiv.org/abs/2605.09630>
- [18] Austin et al. Structured Denoising Diffusion Models in Discrete State-Spaces. <https://arxiv.org/abs/2107.03006>
- [19] Li et al. Diffusion-LM Improves Controllable Text Generation. <https://arxiv.org/abs/2205.14217>
- [20] Lou et al. Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution. <https://arxiv.org/abs/2310.16834>
- [21] Nie et al. Large Language Diffusion Models. <https://arxiv.org/abs/2502.09992>

- [22] Chang et al. MaskGIT: Masked Generative Image Transformer. <https://arxiv.org/abs/2202.04200>
- [23] Rombach et al. High-Resolution Image Synthesis with Latent Diffusion Models. <https://arxiv.org/abs/2112.10752>
- [24] Peebles and Xie. Scalable Diffusion Models with Transformers. <https://arxiv.org/abs/2212.09748>
- [25] Lipman et al. Flow Matching for Generative Modeling. <https://arxiv.org/abs/2210.02747>
- [26] Chen and Lipman. Riemannian Flow Matching on General Geometries. <https://arxiv.org/abs/2302.03660>
- [27] Mathieu and Nickel. Riemannian Continuous Normalizing Flows. <https://arxiv.org/abs/2006.10605>
- [28] Assran et al. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. <https://arxiv.org/abs/2301.08243>
- [29] Velarde and Parra. Recurrent Joint Embedding Predictive Architecture with Recurrent Forward Propagation Learning. <https://arxiv.org/abs/2411.16695>
- [30] Chen et al. Denoising with a Joint-Embedding Predictive Architecture. <https://arxiv.org/abs/2410.03755>
- [31] Hu et al. ELF: Embedded Language Flows. <https://arxiv.org/abs/2605.10938>
- [32] Shao, Meng, and Zhou. Continuous Visual Autoregressive Generation via Score Maximization. <https://arxiv.org/abs/2505.07812>
- [33] Chen et al. VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. <https://arxiv.org/abs/2512.10942>
- [34] Jiang et al. TextLDM: Language Modeling with Continuous Latent Diffusion. <https://arxiv.org/abs/2605.07748>
- [35] Ross, Gordon, and Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. <https://proceedings.mlr.press/v15/ross11a.html>
- [36] Bengio et al. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. https://papers.nips.cc/paper_files/paper/2015/hash/e995f98d56967d946471af29d7bf99f1-Abstract.html
- [37] Ning et al. Input Perturbation Reduces Exposure Bias in Diffusion Models. <https://proceedings.mlr.press/v202/ning23a.html>
- [38] Yan et al. Eye movement guidance in Chinese reading: Is there a preferred viewing location? *Vision Research*, 51(10), 2011. <https://doi.org/10.1016/j.visres.2011.03.004>